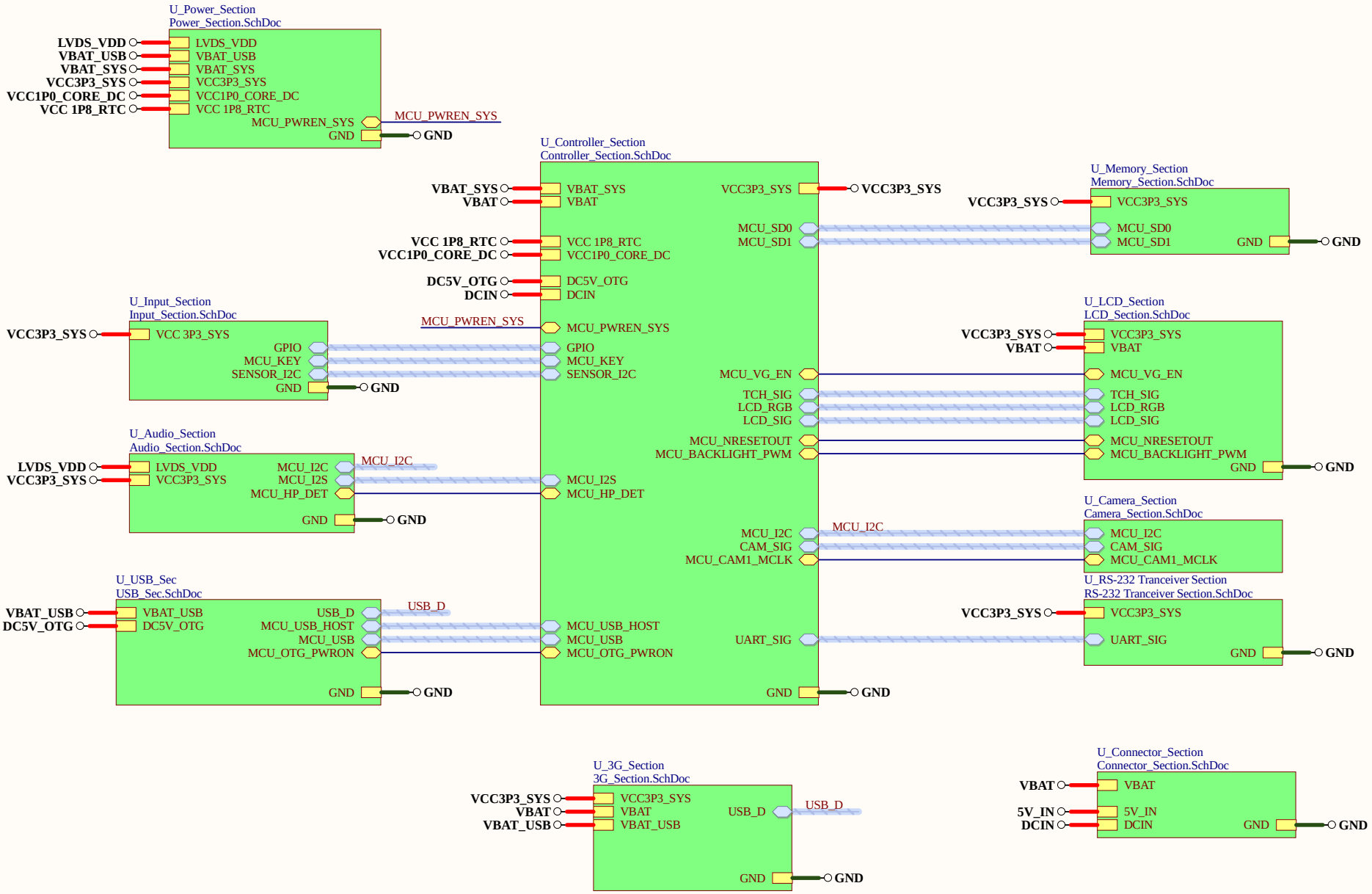
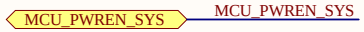
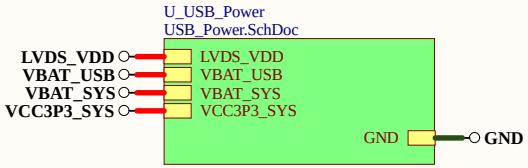
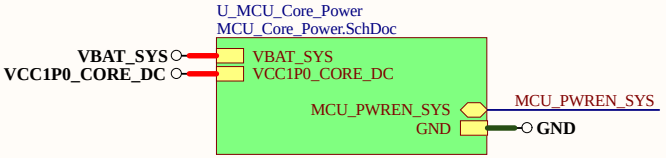
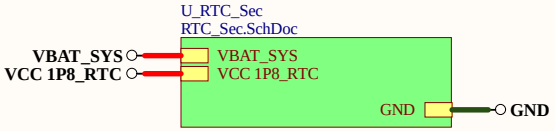
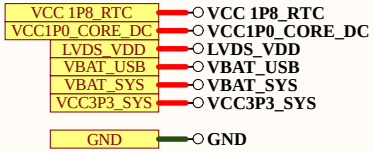

U_Power_Section
Power_Section.SchDoc
LVDS_VDD
VBAT_USB
VBAT_SYS
VCC3P3_SYS
VCC1P0_CORE_DC
VCC 1P8_RTC
MCU_PWREN_SYS
GND
U_Controller_Section
Controller_Section.SchDoc
VBAT_SYS
VBAT
VCC3P3_SYS
VCC 1P8_RTC
VCC1P0_CORE_DC
DC5V_OTG
DCIN
MCU_PWREN_SYS
GPIO
MCU_KEY
SENSOR_I2C
MCU_VG_EN
TCH_SIG
LCD_RGB
LCD_SIG
MCU_NRESETOUT
MCU_BACKLIGHT_PWM
MCU_I2C
MCU_I2S
MCU_HP_DET
CAM_SIG
MCU_CAM1_MCLK
MCU_SD0
MCU_SD1
MCU_USB_HOST
MCU_USB
MCU_OTG_PWRON
UART_SIG
U_Memory_Section
Memory_Section.SchDoc
U_Input_Section
Input_Section.SchDoc
VCC 3P3_SYS
U_LCD_Section
LCD_Section.SchDoc
U_Audio_Section
Audio_Section.SchDoc
U_Camera_Section
Camera_Section.SchDoc
U_USB_Sec
USB_Sec.SchDoc
USB_D
U_RS-232 Tranceiver Section
RS-232 Tranceiver Section.SchDoc
U_3G_Section
3G_Section.SchDoc
U_Connector_Section
Connector_Section.SchDoc
5V_IN
DCIN

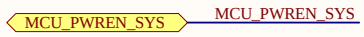

MCU_PWREN_SYS
MCU_PWREN_SYS

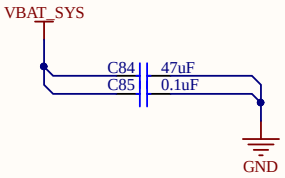

U_USB_Power
USB_Power.SchDoc
LVDS_VDD
VBAT_USB
VBAT_SYS
VCC3P3_SYS
LVDS_VDD
VBAT_USB
VBAT_SYS
VCC3P3_SYS
GND
GND

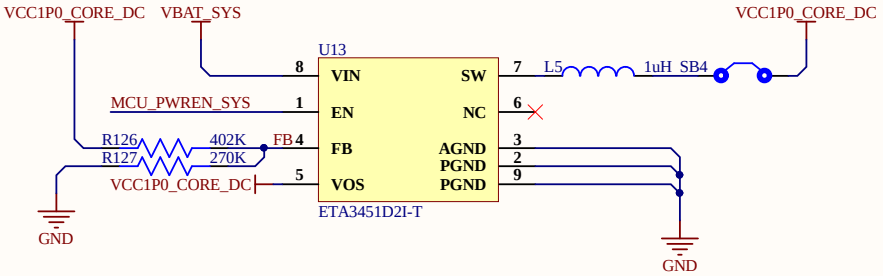

U_MCU_Core_Power
MCU_Core_Power.SchDoc
VBAT_SYS
VCC1P0_CORE_DC
VBAT_SYS
VCC1P0_CORE_DC
MCU_PWREN_SYS
GND
MCU_PWREN_SYS
GND

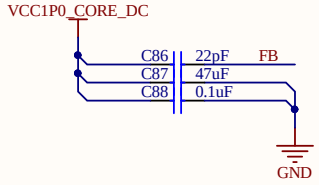

U_RTC_Sec
RTC_Sec.SchDoc
VBAT_SYS
VCC 1P8_RTC
VBAT_SYS
VCC 1P8_RTC
GND
GND

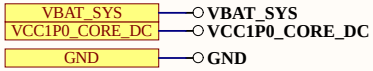

VCC 1P8_RTC
VCC1P0_CORE_DC
LVDS_VDD
VBAT_USB
VBAT_SYS
VCC3P3_SYS
GND
VCC 1P8_RTC
VCC1P0_CORE_DC
LVDS_VDD
VBAT_USB
VBAT_SYS
VCC3P3_SYS
GND

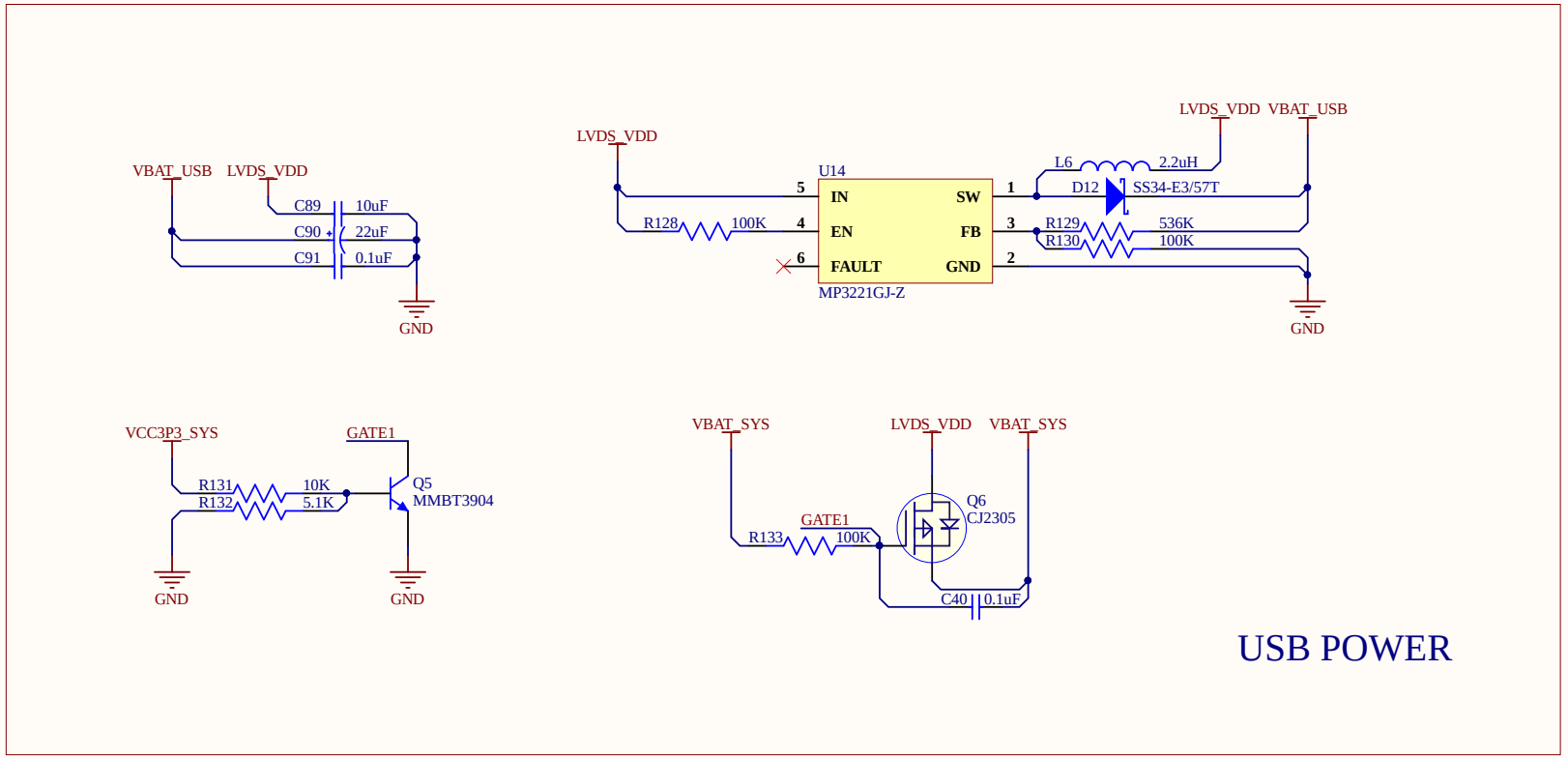

MCU_PWREN_SYS
MCU_PWREN_SYS


VBAT_SYS
C84
47uF
C85
0.1uF
GND

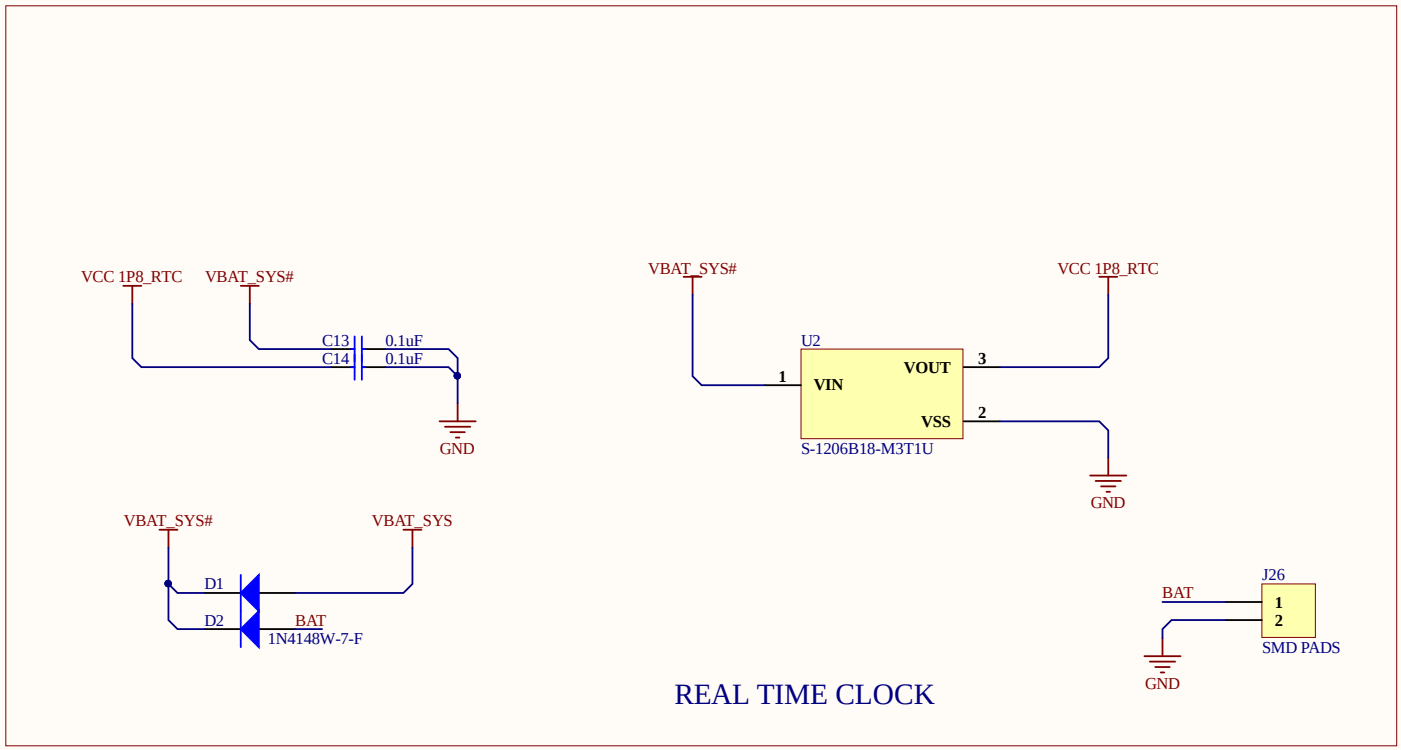

VCC1P0_CORE_DC
VBAT_SYS
U13
8
VIN
SW
7
L5
1uH
SB4
VCC1P0_CORE_DC
MCU_PWREN_SYS
1
EN
NC
6
R126
402K
FB
4
FB
AGND
3
R127
270K
PGND
2
VCC1P0_CORE_DC
5
VOS
PGND
9
ETA3451D2I-T
GND
GND


VCC1P0_CORE_DC
C86
22pF
FB
C87
47uF
C88
0.1uF
GND


VBAT_SYS
VBAT_SYS
VCC1P0_CORE_DC
VCC1P0_CORE_DC
GND
GND



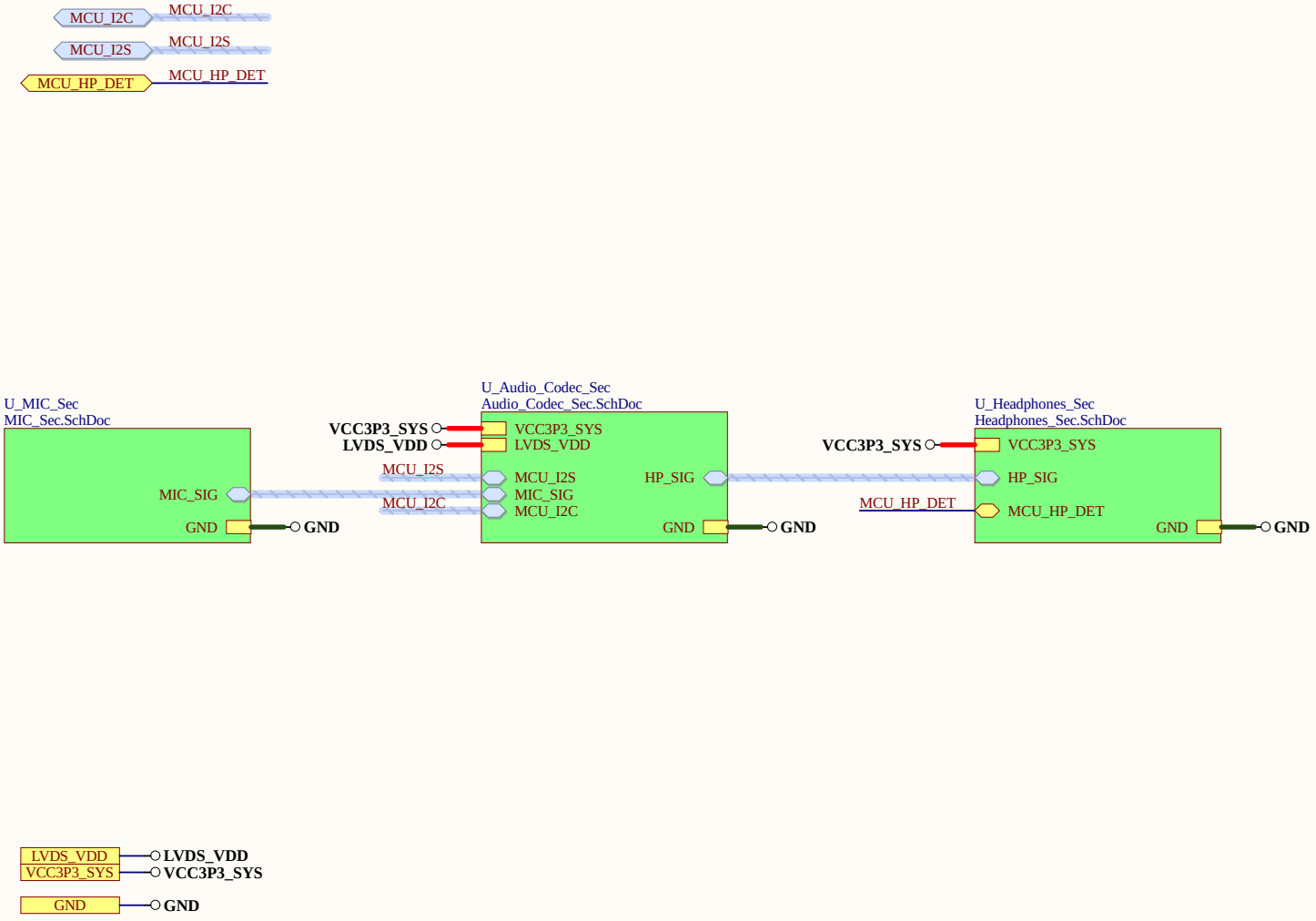

| | |
|---|---|
| LVDS_VDD | **LVDS_VDD** |
| VBAT_USB | **VBAT_USB** |
| VBAT_SYS | **VBAT_SYS** |
| VCC3P3_SYS | **VCC3P3_SYS** |
| GND | **GND** |



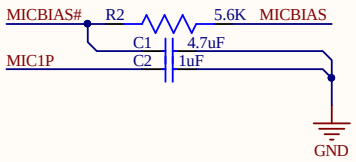

VCC 1P8_RTC → VCC 1P8_RTC

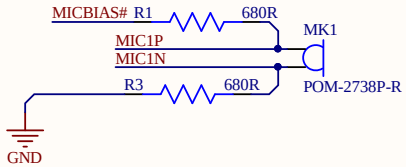
VBAT_SYS → VBAT_SYS

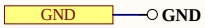
GND → GND

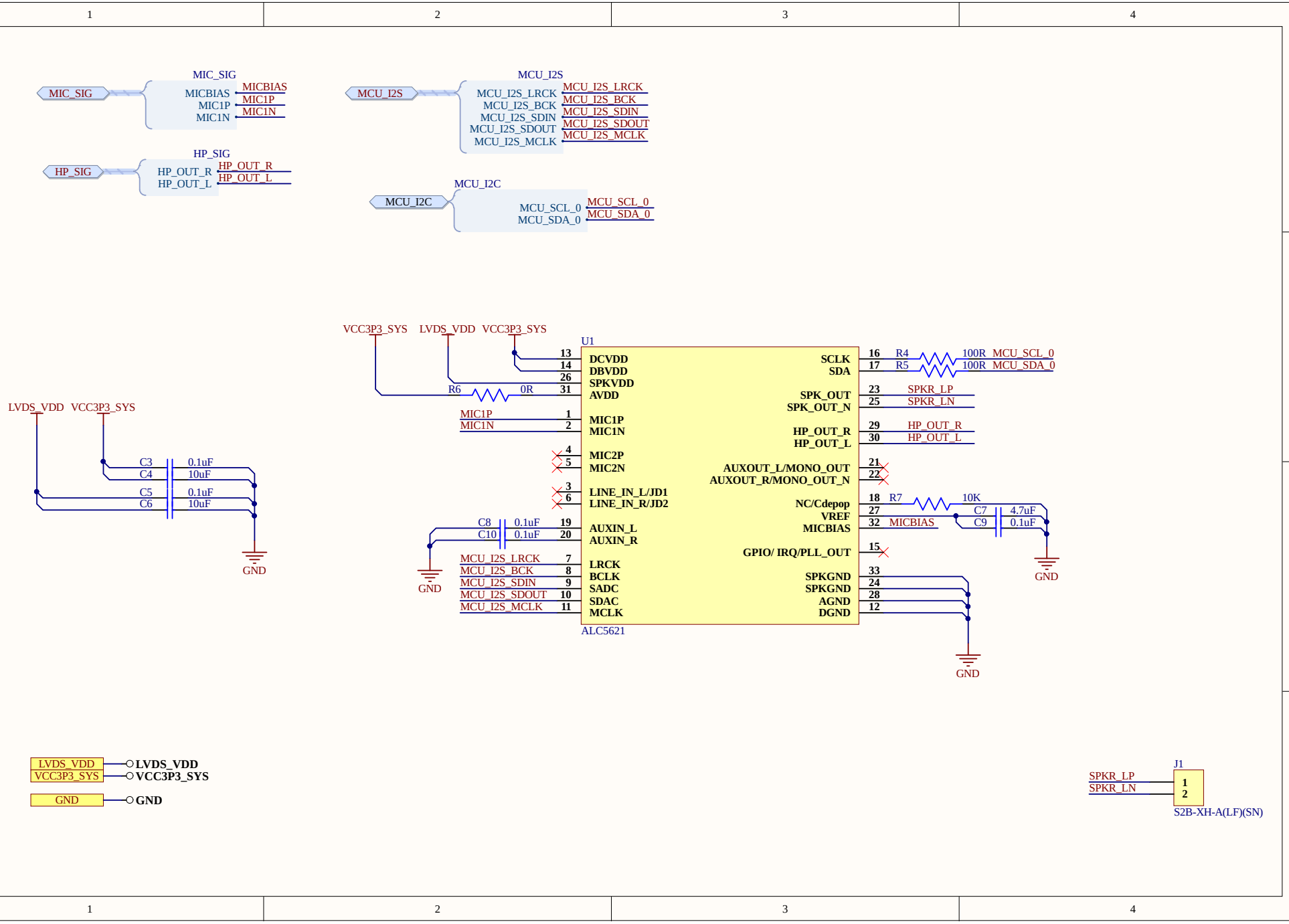

MCU_I2C
MCU_I2C
MCU_I2S
MCU_I2S
MCU_HP_DET
MCU_HP_DET
U_MIC_Sec
MIC_Sec.SchDoc
MIC_SIG
GND
GND
U_Audio_Codec_Sec
Audio_Codec_Sec.SchDoc
VCC3P3_SYS
LVDS_VDD
MCU_I2S
MCU_I2C
VCC3P3_SYS
LVDS_VDD
MCU_I2S
MIC_SIG
MCU_I2C
HP_SIG
GND
GND
U_Headphones_Sec
Headphones_Sec.SchDoc
VCC3P3_SYS
MCU_HP_DET
VCC3P3_SYS
HP_SIG
MCU_HP_DET
GND
GND
LVDS_VDD
VCC3P3_SYS
LVDS_VDD
VCC3P3_SYS
GND
GND

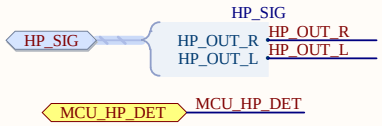

MIC_SIG
MIC_SIG
MICBIAS
MICBIAS
MIC1P
MIC1N
MICBIAS
MIC1P
MIC1N

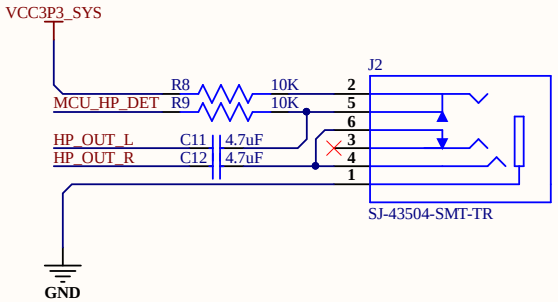

MICBIAS#
R2
5.6K
MICBIAS
C1
4.7uF
MIC1P
C2
1uF
GND

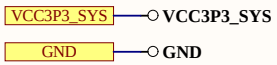

MICBIAS#
R1
680R
MK1
MIC1P
MIC1N
R3
680R
POM-2738P-R
GND

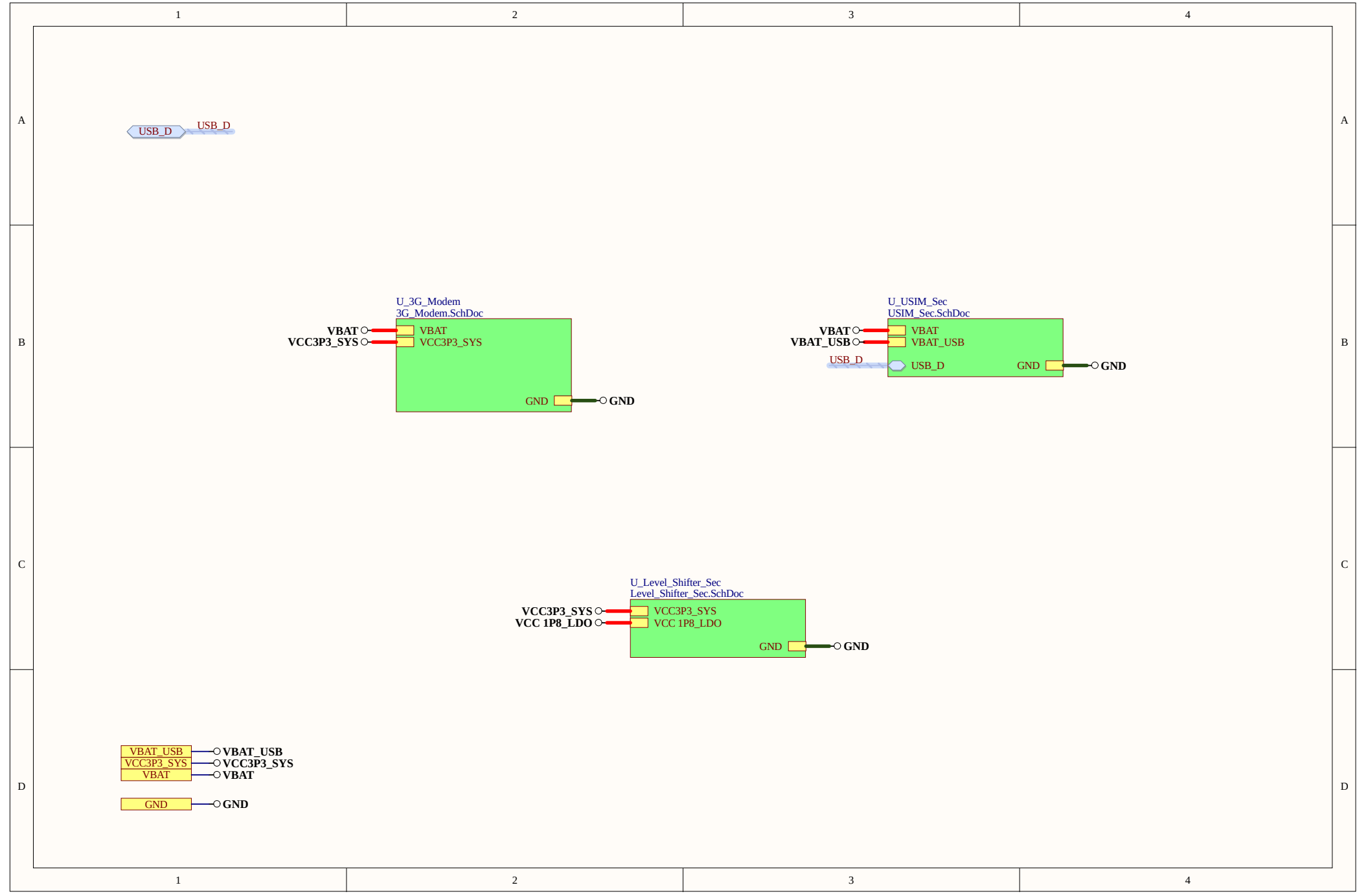

GND
GND

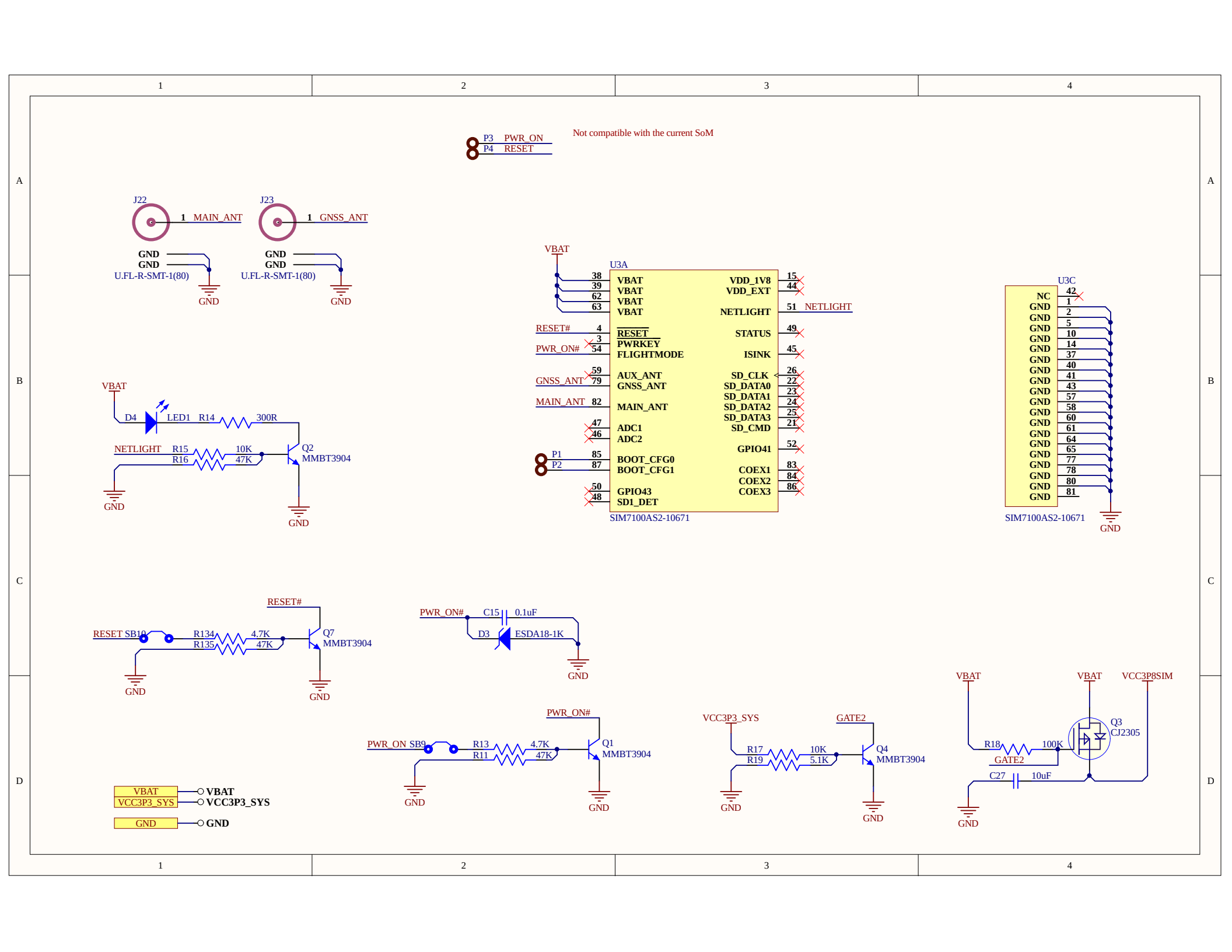

MIC_SIG
MIC_SIG
MICBIAS
MIC1P
MIC1N
MICBIAS
MIC1P
MIC1N
HP_SIG
HP_SIG
HP_OUT_R
HP_OUT_L
HP_OUT_R
HP_OUT_L
MCU_I2S
MCU_I2S
MCU_I2S_LRCK
MCU_I2S_BCK
MCU_I2S_SDIN
MCU_I2S_SDOUT
MCU_I2S_MCLK
MCU_I2S_LRCK
MCU_I2S_BCK
MCU_I2S_SDIN
MCU_I2S_SDOUT
MCU_I2S_MCLK
MCU_I2C
MCU_I2C
MCU_SCL_0
MCU_SDA_0
MCU_SCL_0
MCU_SDA_0
LVDS_VDD
VCC3P3_SYS
C3
C4
C5
C6
0.1uF
10uF
0.1uF
10uF
GND
VCC3P3_SYS
LVDS_VDD
VCC3P3_SYS
U1
R6
0R
MIC1P
MIC1N
MIC2P
MIC2N
LINE_IN_L/JD1
LINE_IN_R/JD2
C8
C10
0.1uF
0.1uF
AUXIN_L
AUXIN_R
MCU_I2S_LRCK
MCU_I2S_BCK
MCU_I2S_SDIN
MCU_I2S_SDOUT
MCU_I2S_MCLK
GND
DCVDD
DBVDD
SPKVDD
AVDD
MIC1P
MIC1N
MIC2P
MIC2N
LINE_IN_L/JD1
LINE_IN_R/JD2
AUXIN_L
AUXIN_R
LRCK
BCLK
SADC
SDAC
MCLK
SCLK
SDA
SPK_OUT
SPK_OUT_N
HP_OUT_R
HP_OUT_L
AUXOUT_L/MONO_OUT
AUXOUT_R/MONO_OUT_N
NC/Cdepop
VREF
MICBIAS
GPIO/ IRQ/PLL_OUT
SPKGND
SPKGND
AGND
DGND
ALC5621
R4
R5
100R
100R
MCU_SCL_0
MCU_SDA_0
SPKR_LP
SPKR_LN
HP_OUT_R
HP_OUT_L
R7
10K
C7
C9
4.7uF
0.1uF
MICBIAS
GND
GND
GND
LVDS_VDD
VCC3P3_SYS
LVDS_VDD
VCC3P3_SYS
GND
GND
J1
SPKR_LP
SPKR_LN
1
2
S2B-XH-A(LF)(SN)

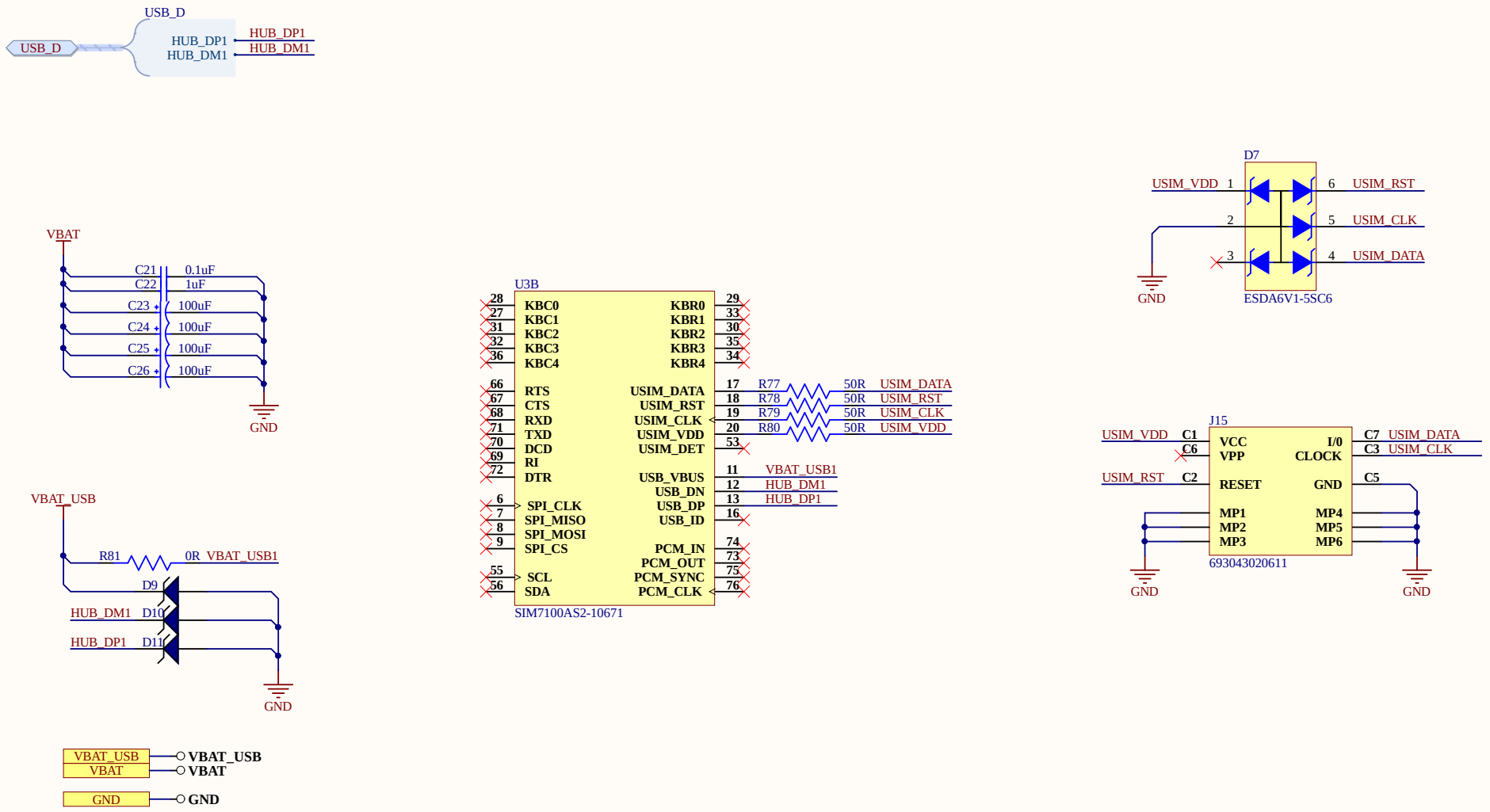

HP_SIG
HP_SIG
HP_OUT_R
HP_OUT_L
HP_OUT_R
HP_OUT_L
MCU_HP_DET
MCU_HP_DET

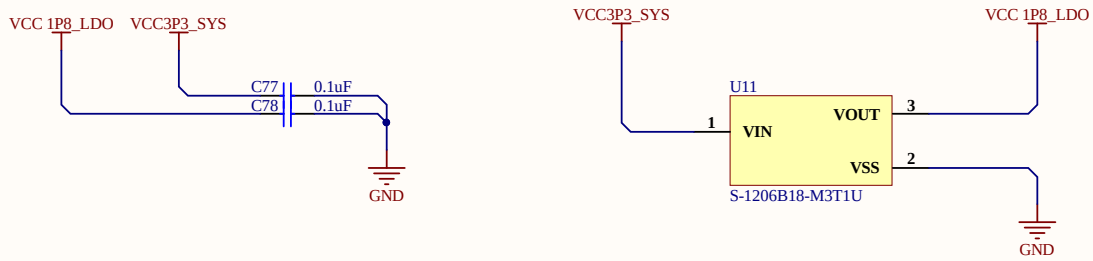

VCC3P3_SYS
J2
R8
10K
MCU_HP_DET
R9
10K
2
5
6
HP_OUT_L
C11
4.7uF
3
HP_OUT_R
C12
4.7uF
4
1
SJ-43504-SMT-TR
GND


VCC3P3_SYS
VCC3P3_SYS
GND
GND

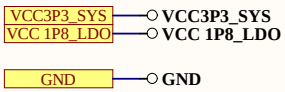

1
2
3
4
A
B
C
D
USB_D
USB_D
U_3G_Modem
3G_Modem.SchDoc
VBAT
VCC3P3_SYS
VBAT
VCC3P3_SYS
GND
GND
U_USIM_Sec
USIM_Sec.SchDoc
VBAT
VBAT_USB
VBAT
VBAT_USB
USB_D
USB_D
GND
GND
U_Level_Shifter_Sec
Level_Shifter_Sec.SchDoc
VCC3P3_SYS
VCC 1P8_LDO
VCC3P3_SYS
VCC 1P8_LDO
GND
GND
VBAT_USB
VCC3P3_SYS
VBAT
VBAT_USB
VCC3P3_SYS
VBAT
GND
GND

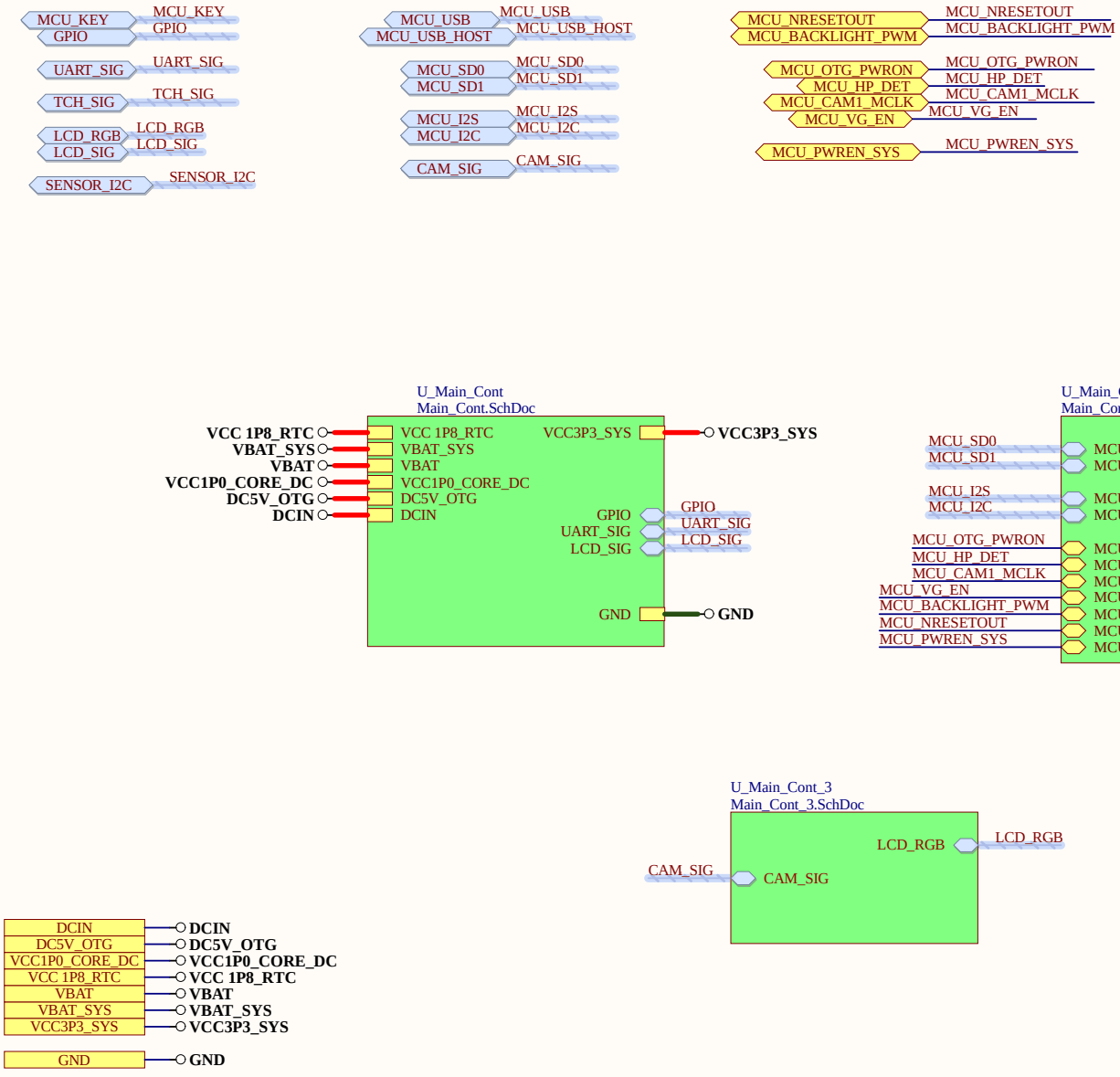

P3 PWR_ON
P4 RESET
Not compatible with the current SoM
J22
1 MAIN_ANT
GND
GND
U.FL-R-SMT-1(80)
J23
1 GNSS_ANT
GND
GND
U.FL-R-SMT-1(80)
VBAT
U3A
38 VBAT
39 VBAT
62 VBAT
63 VBAT
RESET# 4 RESET
3 PWRKEY
PWR_ON# 54 FLIGHTMODE
59 AUX_ANT
GNSS_ANT 79 GNSS_ANT
MAIN_ANT 82 MAIN_ANT
47 ADC1
46 ADC2
P1 85 BOOT_CFG0
P2 87 BOOT_CFG1
50 GPIO43
48 SD1_DET
15 VDD_1V8
44 VDD_EXT
51 NETLIGHT
NETLIGHT
49 STATUS
45 ISINK
26 SD_CLK
22 SD_DATA0
23 SD_DATA1
24 SD_DATA2
25 SD_DATA3
21 SD_CMD
52 GPIO41
83 COEX1
84 COEX2
86 COEX3
SIM7100AS2-10671
U3C
42 NC
1 GND
2 GND
5 GND
10 GND
14 GND
37 GND
40 GND
41 GND
43 GND
57 GND
58 GND
60 GND
61 GND
64 GND
65 GND
77 GND
78 GND
80 GND
81 GND
SIM7100AS2-10671
GND
VBAT
D4 LED1 R14 300R
NETLIGHT R15 10K
R16 47K
Q2 MMBT3904
GND
GND
RESET#
RESET SB10 R134 4.7K
R135 47K
Q7 MMBT3904
GND
GND
PWR_ON# C15 0.1uF
D3 ESDA18-1K
GND
PWR_ON#
PWR_ON SB9 R13 4.7K
R11 47K
Q1 MMBT3904
GND
GND
VCC3P3_SYS
GATE2
R17 10K
R19 5.1K
Q4 MMBT3904
GND
GND
VBAT
VBAT
VCC3P8SIM
Q3 CJ2305
R18 100K
GATE2
C27 10uF
GND
GND
VBAT VBAT
VCC3P3_SYS VCC3P3_SYS
GND GND

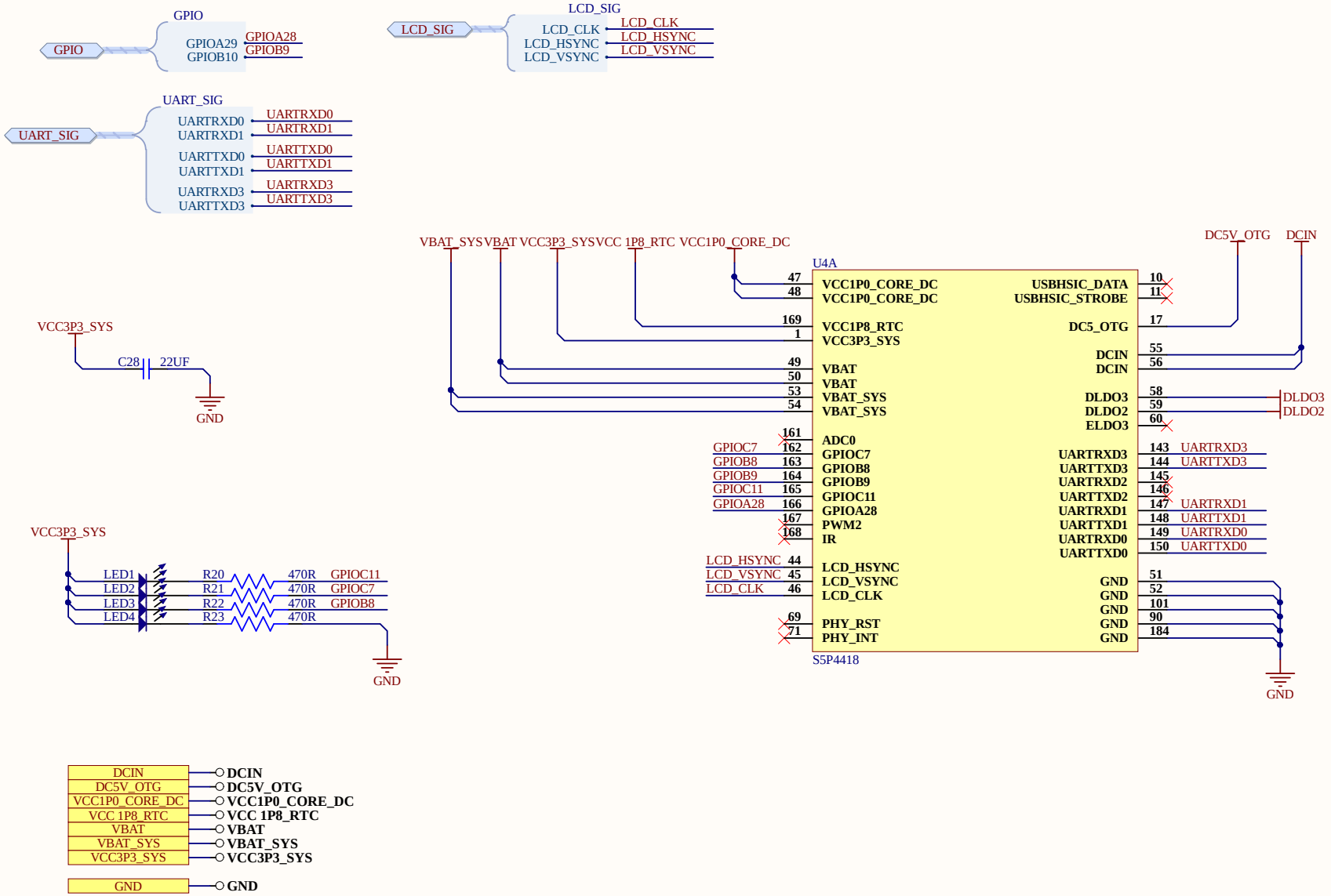

USB_D
USB_D
HUB_DP1
HUB_DM1
HUB_DP1
HUB_DM1
VBAT
C21 0.1uF
C22 1uF
C23 100uF
C24 100uF
C25 100uF
C26 100uF
GND
VBAT_USB
R81 0R VBAT_USB1
D9
HUB_DM1 D10
HUB_DP1 D11
GND
VBAT_USB
VBAT
VBAT_USB
VBAT
GND
GND
U3B
KBC0 KBC1 KBC2 KBC3 KBC4
RTS CTS RXD TXD DCD RI DTR
SPI_CLK SPI_MISO SPI_MOSI SPI_CS
SCL SDA
KBR0 KBR1 KBR2 KBR3 KBR4
USIM_DATA USIM_RST USIM_CLK USIM_VDD USIM_DET
USB_VBUS USB_DN USB_DP USB_ID
PCM_IN PCM_OUT PCM_SYNC PCM_CLK
R77 50R USIM_DATA
R78 50R USIM_RST
R79 50R USIM_CLK
R80 50R USIM_VDD
VBAT_USB1
HUB_DM1
HUB_DP1
SIM7100AS2-10671
D7
USIM_VDD
USIM_RST
USIM_CLK
USIM_DATA
GND
ESDA6V1-5SC6
J15
USIM_VDD C1
C6
USIM_RST C2
VCC VPP RESET GND MP1 MP2 MP3 I/0 CLOCK GND MP4 MP5 MP6
C7 USIM_DATA
C3 USIM_CLK
C5
693043020611
GND
GND

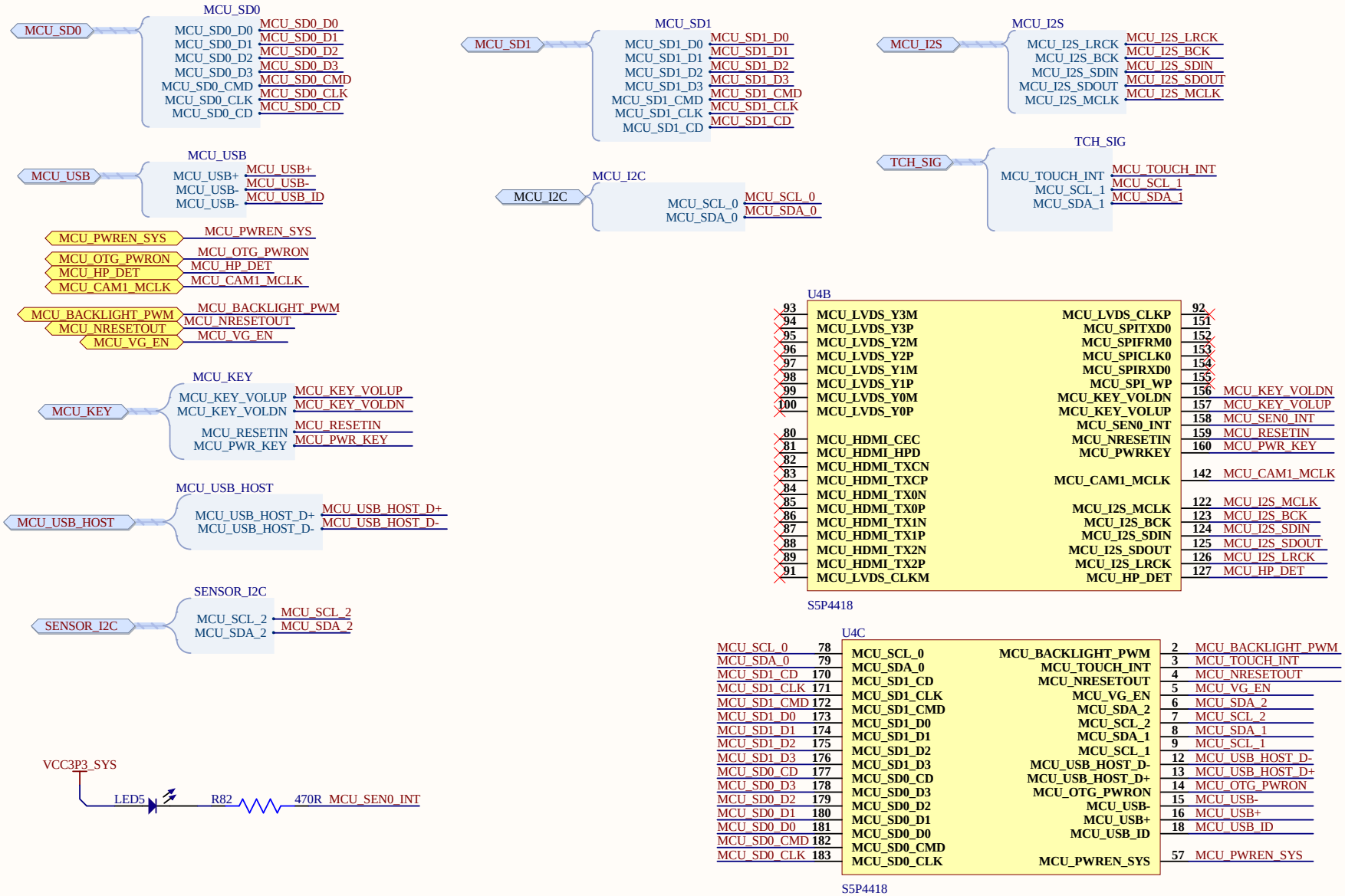


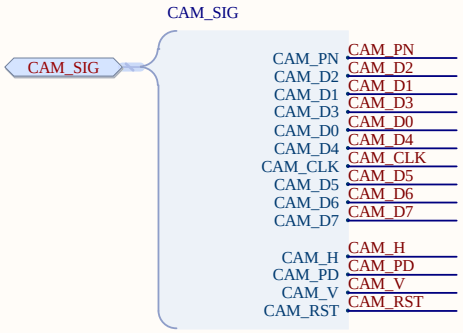

LEVEL SHIFTER  POWER CIRCUIT

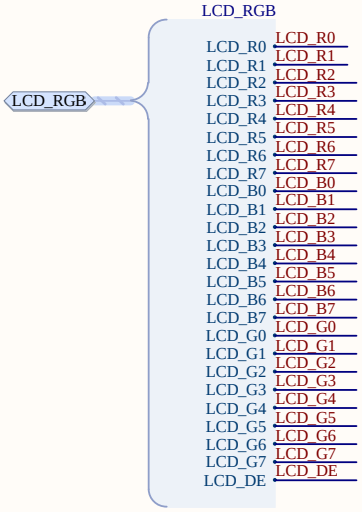


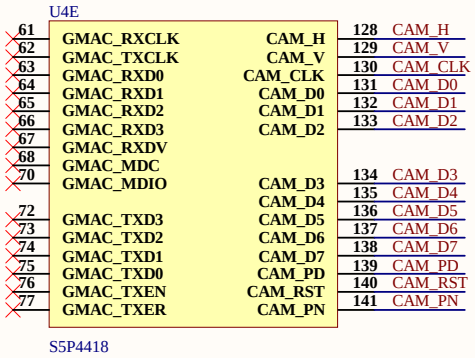

MCU_KEY
GPIO
UART_SIG
TCH_SIG
LCD_RGB
LCD_SIG
SENSOR_I2C
MCU_USB
MCU_USB_HOST
MCU_SD0
MCU_SD1
MCU_I2S
MCU_I2C
CAM_SIG
MCU_NRESETOUT
MCU_BACKLIGHT_PWM
MCU_OTG_PWRON
MCU_HP_DET
MCU_CAM1_MCLK
MCU_VG_EN
MCU_PWREN_SYS
U_Main_Cont
Main_Cont.SchDoc
VCC 1P8_RTC
VBAT_SYS
VBAT
VCC1P0_CORE_DC
DC5V_OTG
DCIN
VCC3P3_SYS
GND
U_Main_Cont_2
Main_Cont_2.SchDoc
U_Main_Cont_3
Main_Cont_3.SchDoc

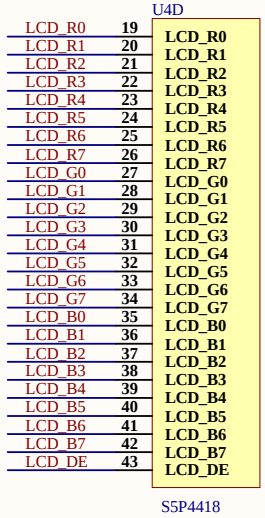

GPIO
GPIO
GPIOA29
GPIOB10
GPIOA28
GPIOB9
LCD_SIG
LCD_SIG
LCD_CLK
LCD_HSYNC
LCD_VSYNC
LCD_CLK
LCD_HSYNC
LCD_VSYNC
UART_SIG
UART_SIG
UARTRXD0
UARTRXD1
UARTTXD0
UARTTXD1
UARTRXD3
UARTTXD3
UARTRXD0
UARTRXD1
UARTTXD0
UARTTXD1
UARTRXD3
UARTTXD3
VBAT_SYS
VBAT
VCC3P3_SYS
VCC 1P8_RTC
VCC1P0_CORE_DC
U4A
47
48
169
1
49
50
53
54
161
162
163
164
165
166
167
168
44
45
46
69
71
VCC1P0_CORE_DC
VCC1P0_CORE_DC
VCC1P8_RTC
VCC3P3_SYS
VBAT
VBAT
VBAT_SYS
VBAT_SYS
ADC0
GPIOC7
GPIOB8
GPIOB9
GPIOC11
GPIOA28
PWM2
IR
LCD_HSYNC
LCD_VSYNC
LCD_CLK
PHY_RST
PHY_INT
USBHSIC_DATA
USBHSIC_STROBE
DC5_OTG
DCIN
DCIN
DLDO3
DLDO2
ELDO3
UARTRXD3
UARTTXD3
UARTRXD2
UARTTXD2
UARTRXD1
UARTTXD1
UARTRXD0
UARTTXD0
GND
GND
GND
GND
GND
10
11
17
55
56
58
59
60
143
144
145
146
147
148
149
150
51
52
101
90
184
S5P4418
DC5V_OTG
DCIN
DLDO3
DLDO2
UARTRXD3
UARTTXD3
UARTRXD1
UARTTXD1
UARTRXD0
UARTTXD0
GPIOC7
GPIOB8
GPIOB9
GPIOC11
GPIOA28
LCD_HSYNC
LCD_VSYNC
LCD_CLK
GND
VCC3P3_SYS
C28
22UF
GND
VCC3P3_SYS
LED1
LED2
LED3
LED4
R20
R21
R22
R23
470R
470R
470R
470R
GPIOC11
GPIOC7
GPIOB8
GND
DCIN
DC5V_OTG
VCC1P0_CORE_DC
VCC 1P8_RTC
VBAT
VBAT_SYS
VCC3P3_SYS
GND
DCIN
DC5V_OTG
VCC1P0_CORE_DC
VCC 1P8_RTC
VBAT
VBAT_SYS
VCC3P3_SYS
GND

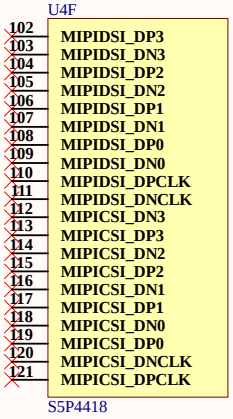

MCU_SD0
MCU_SD0_D0
MCU_SD0_D1
MCU_SD0_D2
MCU_SD0_D3
MCU_SD0_CMD
MCU_SD0_CLK
MCU_SD0_CD
MCU_SD1
MCU_SD1_D0
MCU_SD1_D1
MCU_SD1_D2
MCU_SD1_D3
MCU_SD1_CMD
MCU_SD1_CLK
MCU_SD1_CD
MCU_I2S
MCU_I2S_LRCK
MCU_I2S_BCK
MCU_I2S_SDIN
MCU_I2S_SDOUT
MCU_I2S_MCLK
MCU_USB
MCU_USB+
MCU_USB-
MCU_USB_ID
MCU_I2C
MCU_SCL_0
MCU_SDA_0
TCH_SIG
MCU_TOUCH_INT
MCU_SCL_1
MCU_SDA_1
MCU_PWREN_SYS
MCU_OTG_PWRON
MCU_HP_DET
MCU_CAM1_MCLK
MCU_BACKLIGHT_PWM
MCU_NRESETOUT
MCU_VG_EN
MCU_KEY
MCU_KEY_VOLUP
MCU_KEY_VOLDN
MCU_RESETIN
MCU_PWR_KEY
MCU_USB_HOST
MCU_USB_HOST_D+
MCU_USB_HOST_D-
SENSOR_I2C
MCU_SCL_2
MCU_SDA_2
VCC3P3_SYS
LED5
R82
470R
MCU_SEN0_INT
U4B
S5P4418
U4C
S5P4418

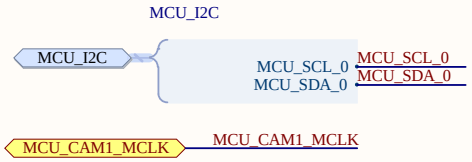

CAM_SIG
CAM_SIG
CAM_PN
CAM_D2
CAM_D1
CAM_D3
CAM_D0
CAM_D4
CAM_CLK
CAM_D5
CAM_D6
CAM_D7
CAM_H
CAM_PD
CAM_V
CAM_RST
CAM_PN
CAM_D2
CAM_D1
CAM_D3
CAM_D0
CAM_D4
CAM_CLK
CAM_D5
CAM_D6
CAM_D7
CAM_H
CAM_PD
CAM_V
CAM_RST

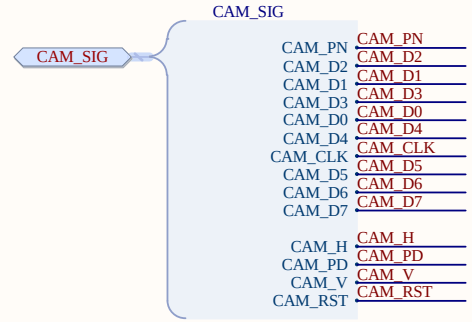

LCD_RGB
LCD_RGB
LCD_R0
LCD_R1
LCD_R2
LCD_R3
LCD_R4
LCD_R5
LCD_R6
LCD_R7
LCD_B0
LCD_B1
LCD_B2
LCD_B3
LCD_B4
LCD_B5
LCD_B6
LCD_B7
LCD_G0
LCD_G1
LCD_G2
LCD_G3
LCD_G4
LCD_G5
LCD_G6
LCD_G7
LCD_DE
LCD_R0
LCD_R1
LCD_R2
LCD_R3
LCD_R4
LCD_R5
LCD_R6
LCD_R7
LCD_B0
LCD_B1
LCD_B2
LCD_B3
LCD_B4
LCD_B5
LCD_B6
LCD_B7
LCD_G0
LCD_G1
LCD_G2
LCD_G3
LCD_G4
LCD_G5
LCD_G6
LCD_G7
LCD_DE

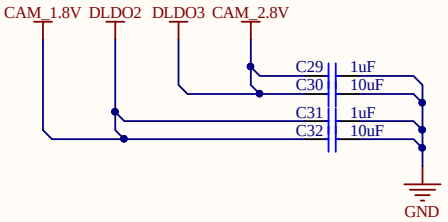

U4E
61 GMAC_RXCLK
62 GMAC_TXCLK
63 GMAC_RXD0
64 GMAC_RXD1
65 GMAC_RXD2
66 GMAC_RXD3
67 GMAC_RXDV
68 GMAC_MDC
70 GMAC_MDIO
72 GMAC_TXD3
73 GMAC_TXD2
74 GMAC_TXD1
75 GMAC_TXD0
76 GMAC_TXEN
77 GMAC_TXER
CAM_H 128 CAM_H
CAM_V 129 CAM_V
CAM_CLK 130 CAM_CLK
CAM_D0 131 CAM_D0
CAM_D1 132 CAM_D1
CAM_D2 133 CAM_D2
CAM_D3 134 CAM_D3
CAM_D4 135 CAM_D4
CAM_D5 136 CAM_D5
CAM_D6 137 CAM_D6
CAM_D7 138 CAM_D7
CAM_PD 139 CAM_PD
CAM_RST 140 CAM_RST
CAM_PN 141 CAM_PN
S5P4418

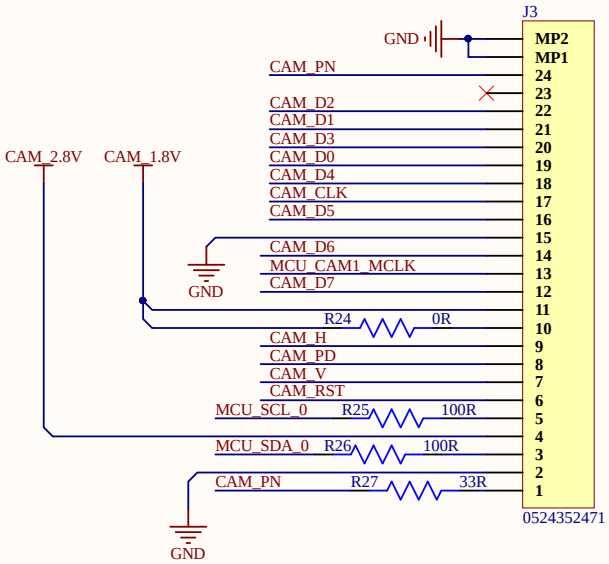

U4D
LCD_R0 19 LCD_R0
LCD_R1 20 LCD_R1
LCD_R2 21 LCD_R2
LCD_R3 22 LCD_R3
LCD_R4 23 LCD_R4
LCD_R5 24 LCD_R5
LCD_R6 25 LCD_R6
LCD_R7 26 LCD_R7
LCD_G0 27 LCD_G0
LCD_G1 28 LCD_G1
LCD_G2 29 LCD_G2
LCD_G3 30 LCD_G3
LCD_G4 31 LCD_G4
LCD_G5 32 LCD_G5
LCD_G6 33 LCD_G6
LCD_G7 34 LCD_G7
LCD_B0 35 LCD_B0
LCD_B1 36 LCD_B1
LCD_B2 37 LCD_B2
LCD_B3 38 LCD_B3
LCD_B4 39 LCD_B4
LCD_B5 40 LCD_B5
LCD_B6 41 LCD_B6
LCD_B7 42 LCD_B7
LCD_DE 43 LCD_DE
S5P4418

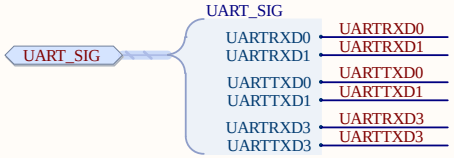

U4F
102 MIPIDSI_DP3
103 MIPIDSI_DN3
104 MIPIDSI_DP2
105 MIPIDSI_DN2
106 MIPIDSI_DP1
107 MIPIDSI_DN1
108 MIPIDSI_DP0
109 MIPIDSI_DN0
110 MIPIDSI_DPCLK
111 MIPIDSI_DNCLK
112 MIPICSI_DN3
113 MIPICSI_DP3
114 MIPICSI_DN2
115 MIPICSI_DP2
116 MIPICSI_DN1
117 MIPICSI_DP1
118 MIPICSI_DN0
119 MIPICSI_DP0
120 MIPICSI_DNCLK
121 MIPICSI_DPCLK
S5P4418

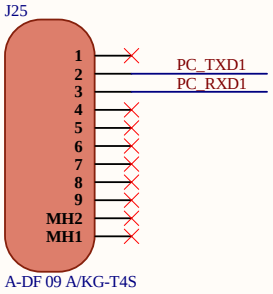

MCU_I2C
MCU_I2C
MCU_SCL_0
MCU_SDA_0
MCU_SCL_0
MCU_SDA_0
MCU_CAM1_MCLK
MCU_CAM1_MCLK

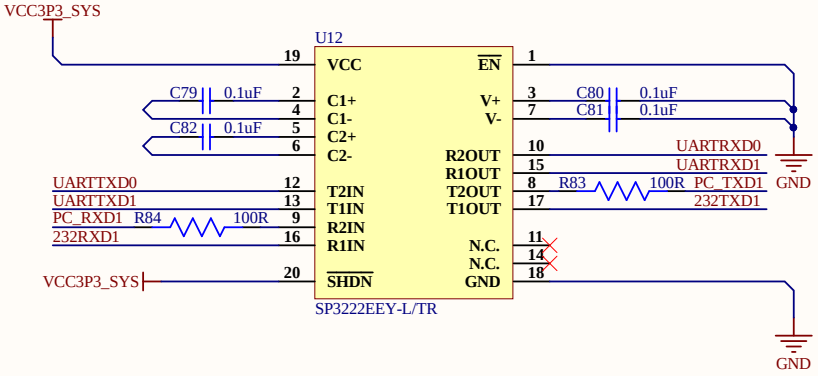

CAM_SIG
CAM_SIG
CAM_PN
CAM_D2
CAM_D1
CAM_D3
CAM_D0
CAM_D4
CAM_CLK
CAM_D5
CAM_D6
CAM_D7
CAM_H
CAM_PD
CAM_V
CAM_RST
CAM_PN
CAM_D2
CAM_D1
CAM_D3
CAM_D0
CAM_D4
CAM_CLK
CAM_D5
CAM_D6
CAM_D7
CAM_H
CAM_PD
CAM_V
CAM_RST

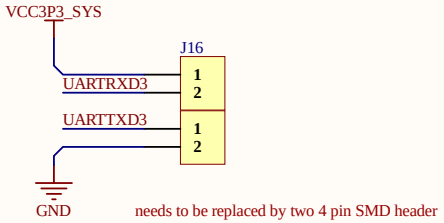

CAM_1.8V
DLDO2
DLDO3
CAM_2.8V
C29
1uF
C30
10uF
C31
1uF
C32
10uF
GND

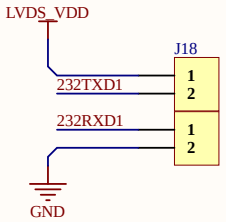

J3
GND
CAM_PN
CAM_D2
CAM_D1
CAM_D3
CAM_D0
CAM_D4
CAM_CLK
CAM_D5
CAM_D6
MCU_CAM1_MCLK
CAM_D7
CAM_H
CAM_PD
CAM_V
CAM_RST
MP2
MP1
24
23
22
21
20
19
18
17
16
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0524352471
CAM_2.8V
CAM_1.8V
GND
R24
0R
MCU_SCL_0
R25
100R
MCU_SDA_0
R26
100R
CAM_PN
R27
33R
GND

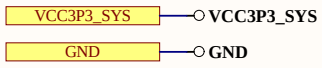

UART_SIG
UART_SIG
UARTRXD0
UARTRXD1
UARTTXD0
UARTTXD1
UARTRXD3
UARTTXD3
UARTRXD0
UARTRXD1
UARTTXD0
UARTTXD1
UARTRXD3
UARTTXD3

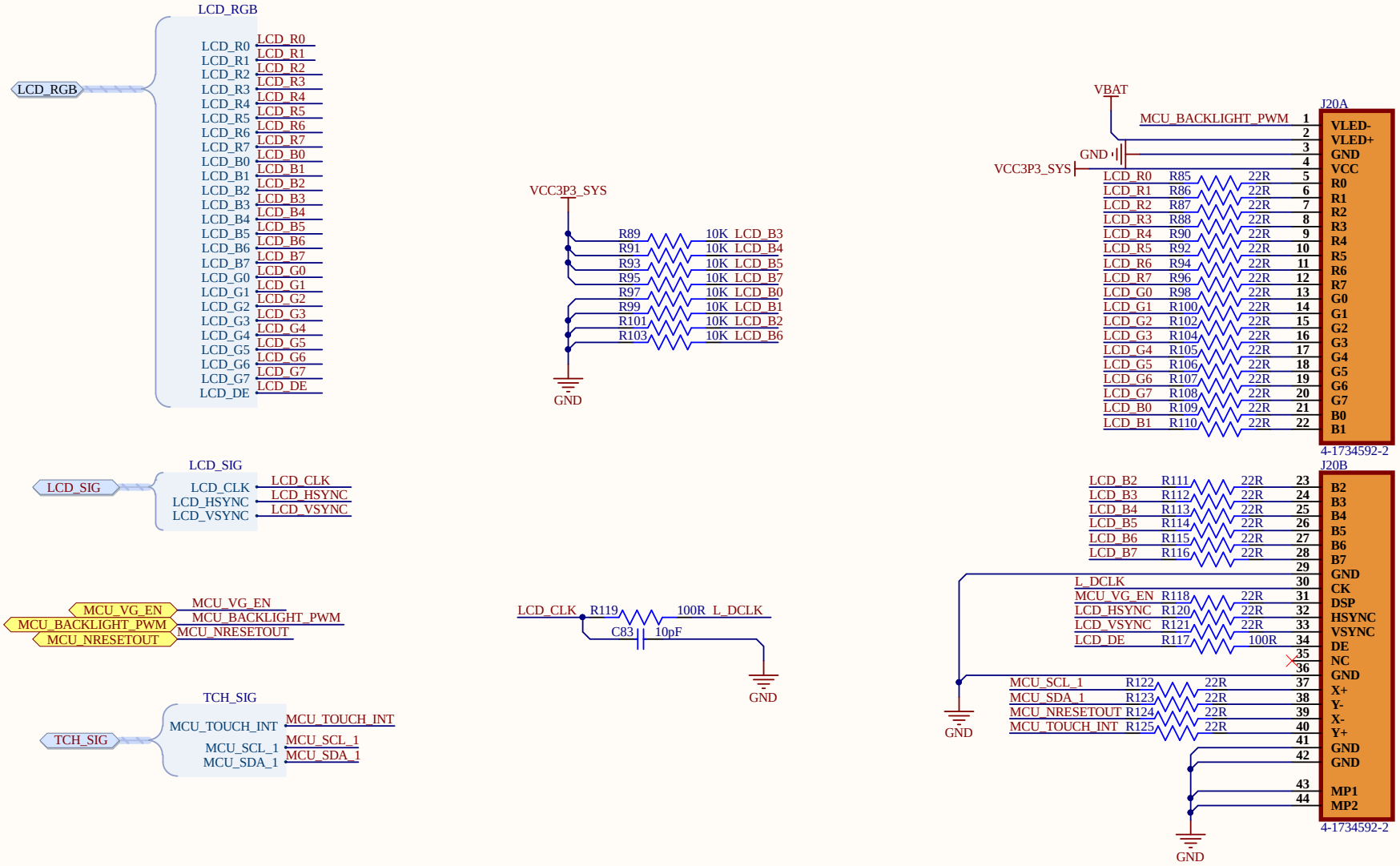

J25
1
2
3
4
5
6
7
8
9
MH2
MH1
PC_TXD1
PC_RXD1
A-DF 09 A/KG-T4S

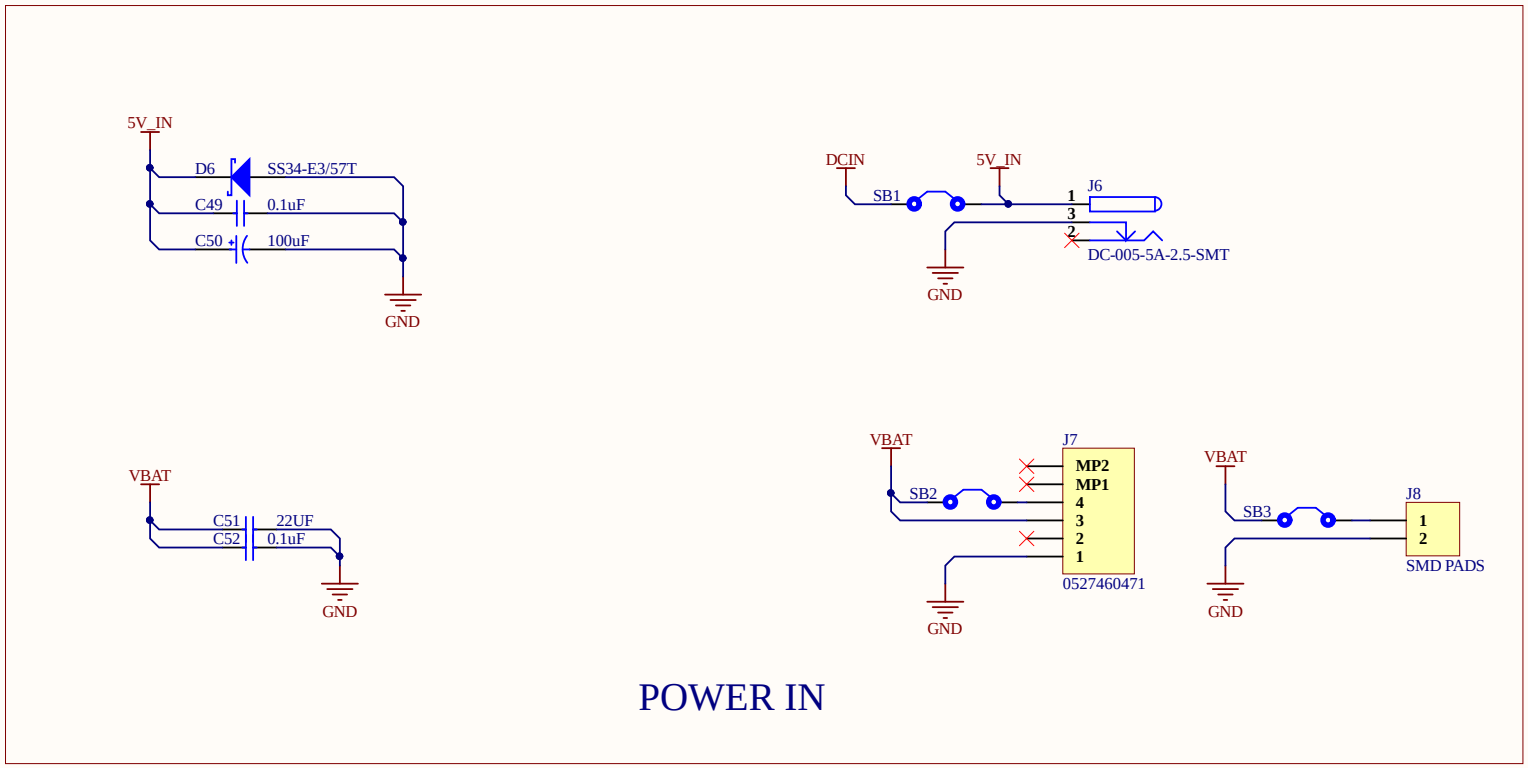

VCC3P3_SYS
U12
19
VCC
EN
1
C79
0.1uF
2
C1+
V+
3
C80
0.1uF
4
C1-
V-
7
C81
0.1uF
C82
0.1uF
5
C2+
6
C2-
R2OUT
10
UARTRXD0
R1OUT
15
UARTRXD1
GND
UARTTXD0
12
T2IN
T2OUT
8
R83
100R
PC_TXD1
UARTTXD1
13
T1IN
T1OUT
17
232TXD1
PC_RXD1
R84
100R
9
R2IN
232RXD1
16
R1IN
N.C.
11
N.C.
14
VCC3P3_SYS
20
SHDN
GND
18
GND
SP3222EEY-L/TR
GND


VCC3P3_SYS
J16
UARTRXD3
1
2
UARTTXD3
1
2
GND
GND
needs to be replaced by two 4 pin SMD header


LVDS_VDD
J18
232TXD1
1
2
232RXD1
1
2
GND

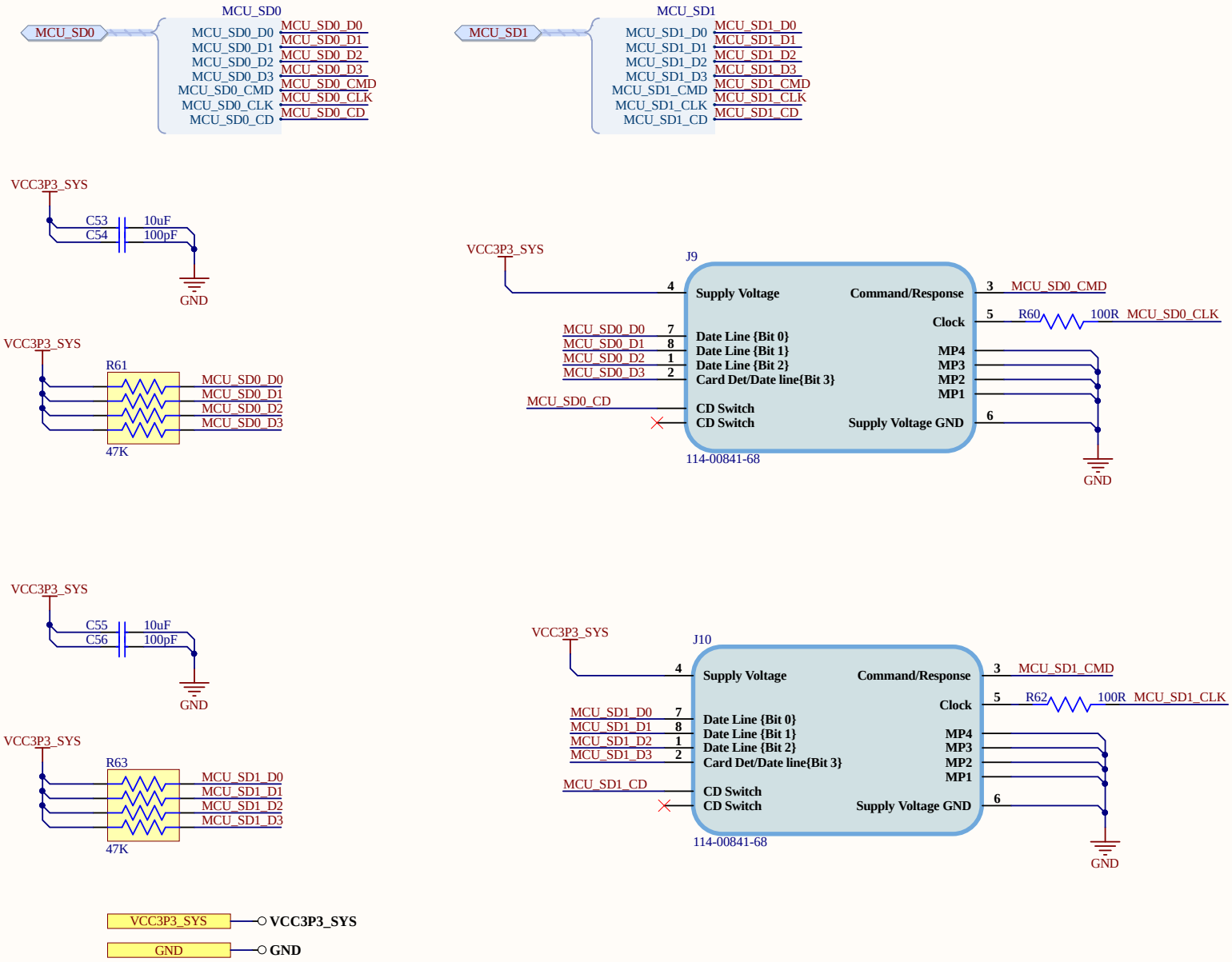

VCC3P3_SYS
VCC3P3_SYS
GND
GND

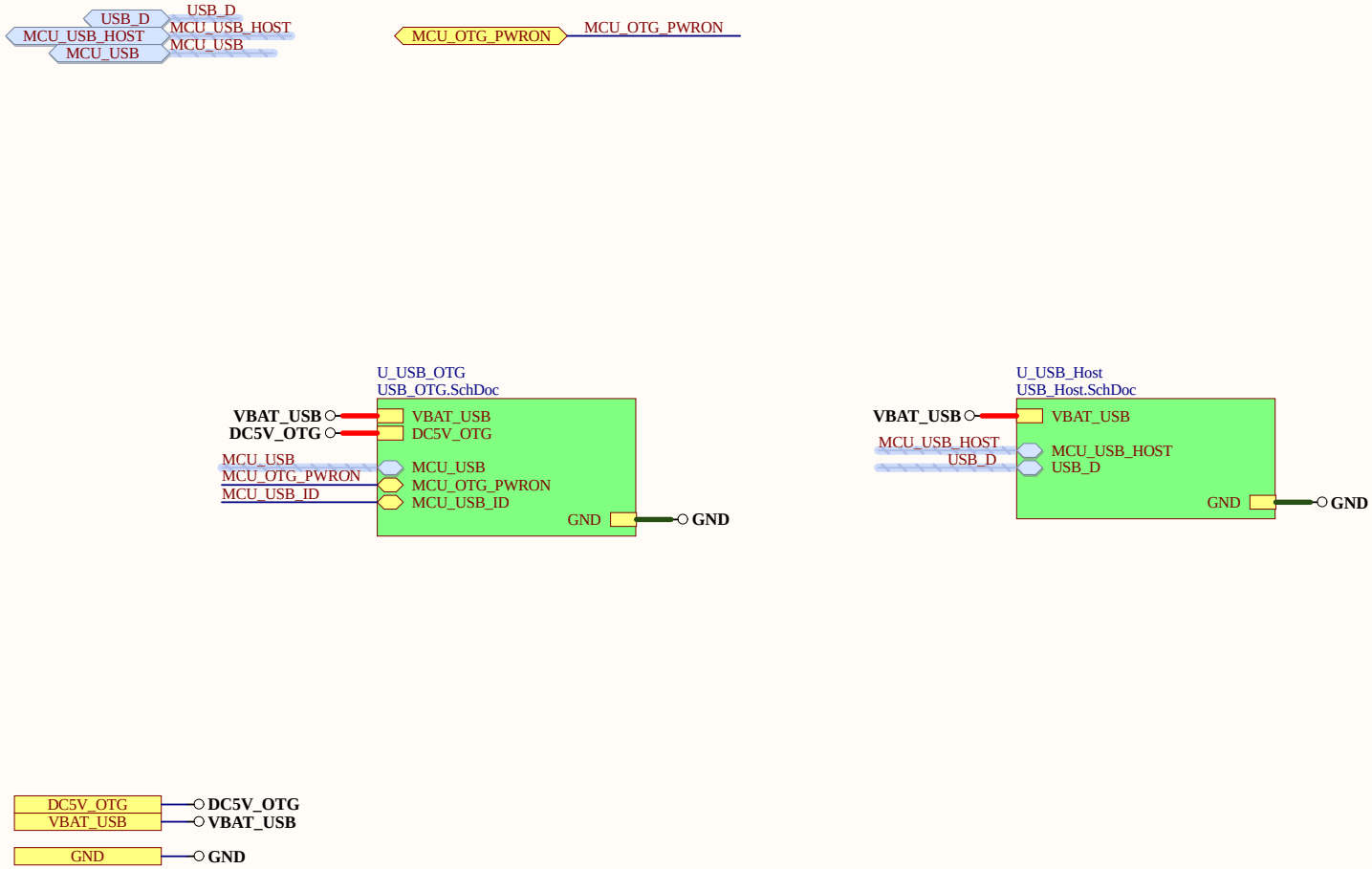

LCD_RGB
LCD_RGB
LCD_R0
LCD_R1
LCD_R2
LCD_R3
LCD_R4
LCD_R5
LCD_R6
LCD_R7
LCD_B0
LCD_B1
LCD_B2
LCD_B3
LCD_B4
LCD_B5
LCD_B6
LCD_B7
LCD_G0
LCD_G1
LCD_G2
LCD_G3
LCD_G4
LCD_G5
LCD_G6
LCD_G7
LCD_DE
LCD_SIG
LCD_SIG
LCD_CLK
LCD_HSYNC
LCD_VSYNC
MCU_VG_EN
MCU_BACKLIGHT_PWM
MCU_NRESETOUT
TCH_SIG
TCH_SIG
MCU_TOUCH_INT
MCU_SCL_1
MCU_SDA_1
VCC3P3_SYS
VBAT
GND
VCC3P3_SYS
R89 10K LCD_B3
R91 10K LCD_B4
R93 10K LCD_B5
R95 10K LCD_B7
R97 10K LCD_B0
R99 10K LCD_B1
R101 10K LCD_B2
R103 10K LCD_B6
GND
LCD_CLK R119 100R L_DCLK
C83 10pF
GND
VBAT
GND
VCC3P3_SYS
J20A
MCU_BACKLIGHT_PWM 1
2
3
4
LCD_R0 R85 22R 5
LCD_R1 R86 22R 6
LCD_R2 R87 22R 7
LCD_R3 R88 22R 8
LCD_R4 R90 22R 9
LCD_R5 R92 22R 10
LCD_R6 R94 22R 11
LCD_R7 R96 22R 12
LCD_G0 R98 22R 13
LCD_G1 R100 22R 14
LCD_G2 R102 22R 15
LCD_G3 R104 22R 16
LCD_G4 R105 22R 17
LCD_G5 R106 22R 18
LCD_G6 R107 22R 19
LCD_G7 R108 22R 20
LCD_B0 R109 22R 21
LCD_B1 R110 22R 22
VLED-
VLED+
GND
VCC
R0
R1
R2
R3
R4
R5
R6
R7
G0
G1
G2
G3
G4
G5
G6
G7
B0
B1
4-1734592-2
J20B
LCD_B2 R111 22R 23
LCD_B3 R112 22R 24
LCD_B4 R113 22R 25
LCD_B5 R114 22R 26
LCD_B6 R115 22R 27
LCD_B7 R116 22R 28
29
L_DCLK 30
MCU_VG_EN R118 22R 31
LCD_HSYNC R120 22R 32
LCD_VSYNC R121 22R 33
LCD_DE R117 100R 34
35
36
MCU_SCL_1 R122 22R 37
MCU_SDA_1 R123 22R 38
MCU_NRESETOUT R124 22R 39
MCU_TOUCH_INT R125 22R 40
41
42
43
44
B2
B3
B4
B5
B6
B7
GND
CK
DSP
HSYNC
VSYNC
DE
NC
GND
X+
Y-
X-
Y+
GND
GND
MP1
MP2
4-1734592-2
GND
GND

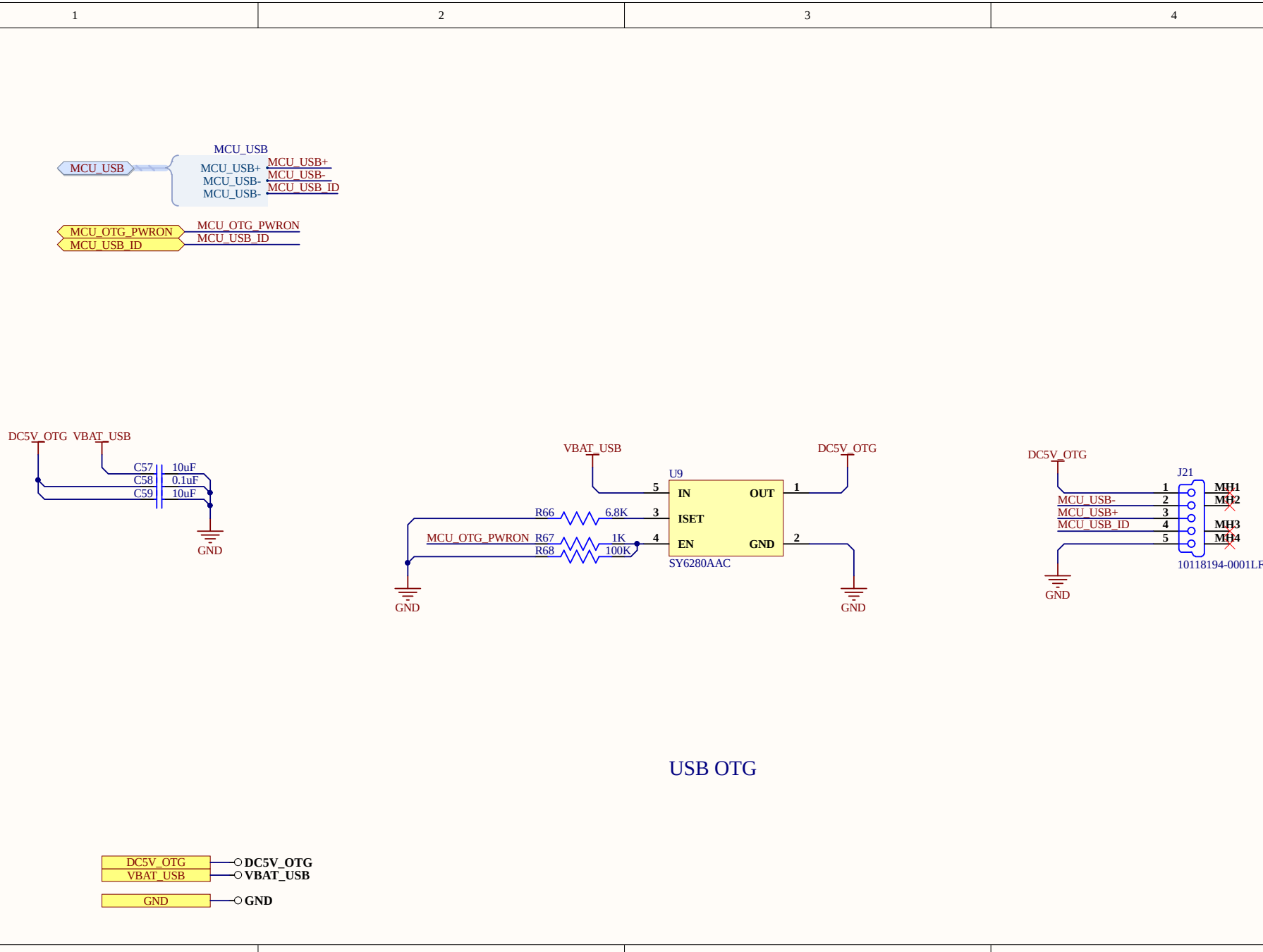


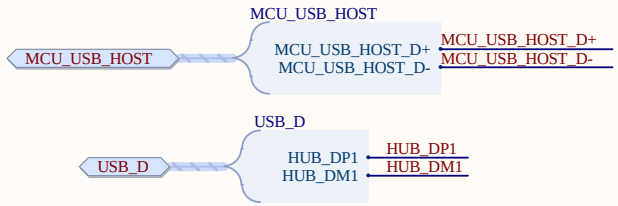

DCIN
5V_IN
VBAT
GND

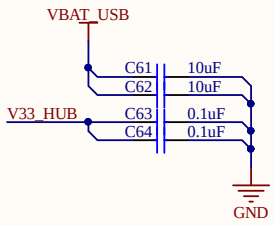
**DCIN**
**5V_IN**
**VBAT**
**GND**

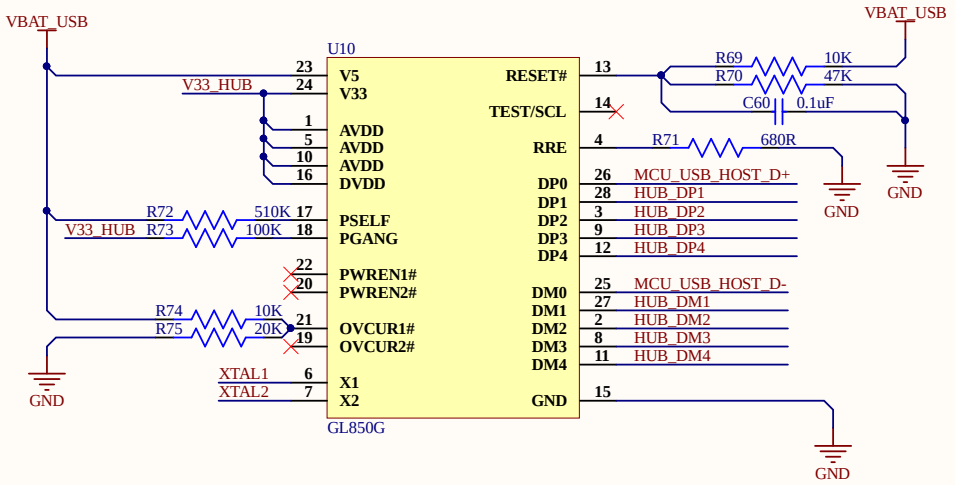

MCU_SD0
MCU_SD0
MCU_SD0_D0
MCU_SD0_D1
MCU_SD0_D2
MCU_SD0_D3
MCU_SD0_CMD
MCU_SD0_CLK
MCU_SD0_CD
MCU_SD0_D0
MCU_SD0_D1
MCU_SD0_D2
MCU_SD0_D3
MCU_SD0_CMD
MCU_SD0_CLK
MCU_SD0_CD
MCU_SD1
MCU_SD1
MCU_SD1_D0
MCU_SD1_D1
MCU_SD1_D2
MCU_SD1_D3
MCU_SD1_CMD
MCU_SD1_CLK
MCU_SD1_CD
MCU_SD1_D0
MCU_SD1_D1
MCU_SD1_D2
MCU_SD1_D3
MCU_SD1_CMD
MCU_SD1_CLK
MCU_SD1_CD
VCC3P3_SYS
C53
10uF
C54
100pF
GND
VCC3P3_SYS
R61
MCU_SD0_D0
MCU_SD0_D1
MCU_SD0_D2
MCU_SD0_D3
47K
VCC3P3_SYS
J9
4
Supply Voltage
Command/Response
3
MCU_SD0_CMD
Clock
5
R60
100R
MCU_SD0_CLK
MCU_SD0_D0
7
Date Line {Bit 0}
MCU_SD0_D1
8
Date Line {Bit 1}
MCU_SD0_D2
1
Date Line {Bit 2}
MCU_SD0_D3
2
Card Det/Date line{Bit 3}
MP4
MP3
MP2
MP1
MCU_SD0_CD
CD Switch
CD Switch
Supply Voltage GND
6
114-00841-68
GND
VCC3P3_SYS
C55
10uF
C56
100pF
GND
VCC3P3_SYS
R63
MCU_SD1_D0
MCU_SD1_D1
MCU_SD1_D2
MCU_SD1_D3
47K
VCC3P3_SYS
J10
4
Supply Voltage
Command/Response
3
MCU_SD1_CMD
Clock
5
R62
100R
MCU_SD1_CLK
MCU_SD1_D0
7
Date Line {Bit 0}
MCU_SD1_D1
8
Date Line {Bit 1}
MCU_SD1_D2
1
Date Line {Bit 2}
MCU_SD1_D3
2
Card Det/Date line{Bit 3}
MP4
MP3
MP2
MP1
MCU_SD1_CD
CD Switch
CD Switch
Supply Voltage GND
6
114-00841-68
GND
VCC3P3_SYS
VCC3P3_SYS
GND
GND

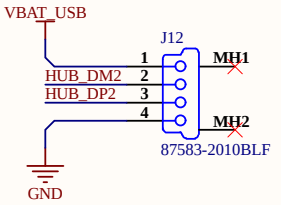

USB_D
MCU_USB_HOST
MCU_USB
USB_D
MCU_USB_HOST
MCU_USB
MCU_OTG_PWRON
MCU_OTG_PWRON
U_USB_OTG
USB_OTG.SchDoc
VBAT_USB
DC5V_OTG
MCU_USB
MCU_OTG_PWRON
MCU_USB_ID
GND
VBAT_USB
DC5V_OTG
MCU_USB
MCU_OTG_PWRON
MCU_USB_ID
GND
GND
U_USB_Host
USB_Host.SchDoc
VBAT_USB
MCU_USB_HOST
USB_D
VBAT_USB
MCU_USB_HOST
USB_D
GND
GND
DC5V_OTG
VBAT_USB
DC5V_OTG
VBAT_USB
GND
GND

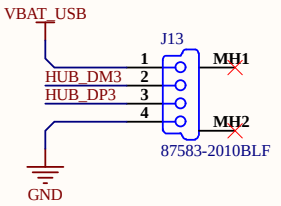

MCU_USB
MCU_USB
MCU_USB+
MCU_USB-
MCU_USB-
MCU_USB+
MCU_USB-
MCU_USB_ID
MCU_OTG_PWRON
MCU_USB_ID
MCU_OTG_PWRON
MCU_USB_ID
DC5V_OTG
VBAT_USB
C57
C58
C59
10uF
0.1uF
10uF
GND
VBAT_USB
DC5V_OTG
U9
5
IN
OUT
1
R66
6.8K
3
ISET
MCU_OTG_PWRON
R67
R68
1K
100K
4
EN
GND
2
SY6280AAC
GND
GND
DC5V_OTG
J21
1
2
3
4
5
MH1
MH2
MH3
MH4
MCU_USB-
MCU_USB+
MCU_USB_ID
10118194-0001LF
GND
USB OTG
DC5V_OTG
VBAT_USB
DC5V_OTG
VBAT_USB
GND
GND

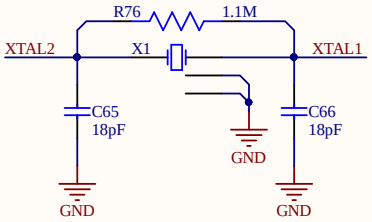

MCU_USB_HOST
MCU_USB_HOST
MCU_USB_HOST_D+
MCU_USB_HOST_D-
MCU_USB_HOST_D+
MCU_USB_HOST_D-
USB_D
USB_D
HUB_DP1
HUB_DM1
HUB_DP1
HUB_DM1


VBAT_USB
C61 10uF
C62 10uF
V33_HUB
C63 0.1uF
C64 0.1uF
GND

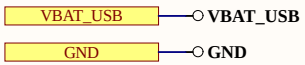

VBAT_USB
U10
V33_HUB
23 V5
24 V33
1 AVDD
5 AVDD
10 AVDD
16 DVDD
R72 510K
V33_HUB R73 100K
17 PSELF
18 PGANG
22 PWREN1#
20 PWREN2#
R74 10K
R75 20K
21 OVCUR1#
19 OVCUR2#
XTAL1 6 X1
XTAL2 7 X2
GND
GL850G
RESET# 13
TEST/SCL 14
RRE 4
DP0 26 MCU_USB_HOST_D+
DP1 28 HUB_DP1
DP2 3 HUB_DP2
DP3 9 HUB_DP3
DP4 12 HUB_DP4
DM0 25 MCU_USB_HOST_D-
DM1 27 HUB_DM1
DM2 2 HUB_DM2
DM3 8 HUB_DM3
DM4 11 HUB_DM4
GND 15
VBAT_USB
R69 10K
R70 47K
C60 0.1uF
R71 680R
GND
GND

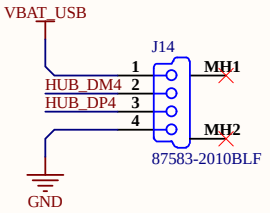

VBAT_USB
J12
1 MH1
HUB_DM2 2
HUB_DP2 3
4 MH2
87583-2010BLF
GND

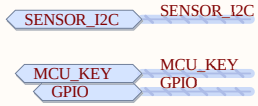

VBAT_USB
J13
1 MH1
HUB_DM3 2
HUB_DP3 3
4 MH2
87583-2010BLF
GND

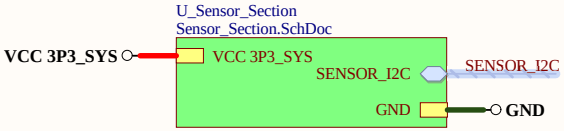

R76 1.1M
XTAL2
X1
XTAL1
C65 18pF
C66 18pF
GND
GND
GND

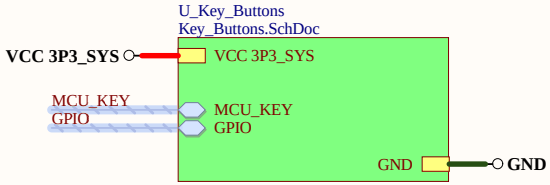
# USB HOST

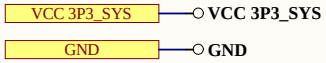

VBAT_USB
VBAT_USB
GND
GND

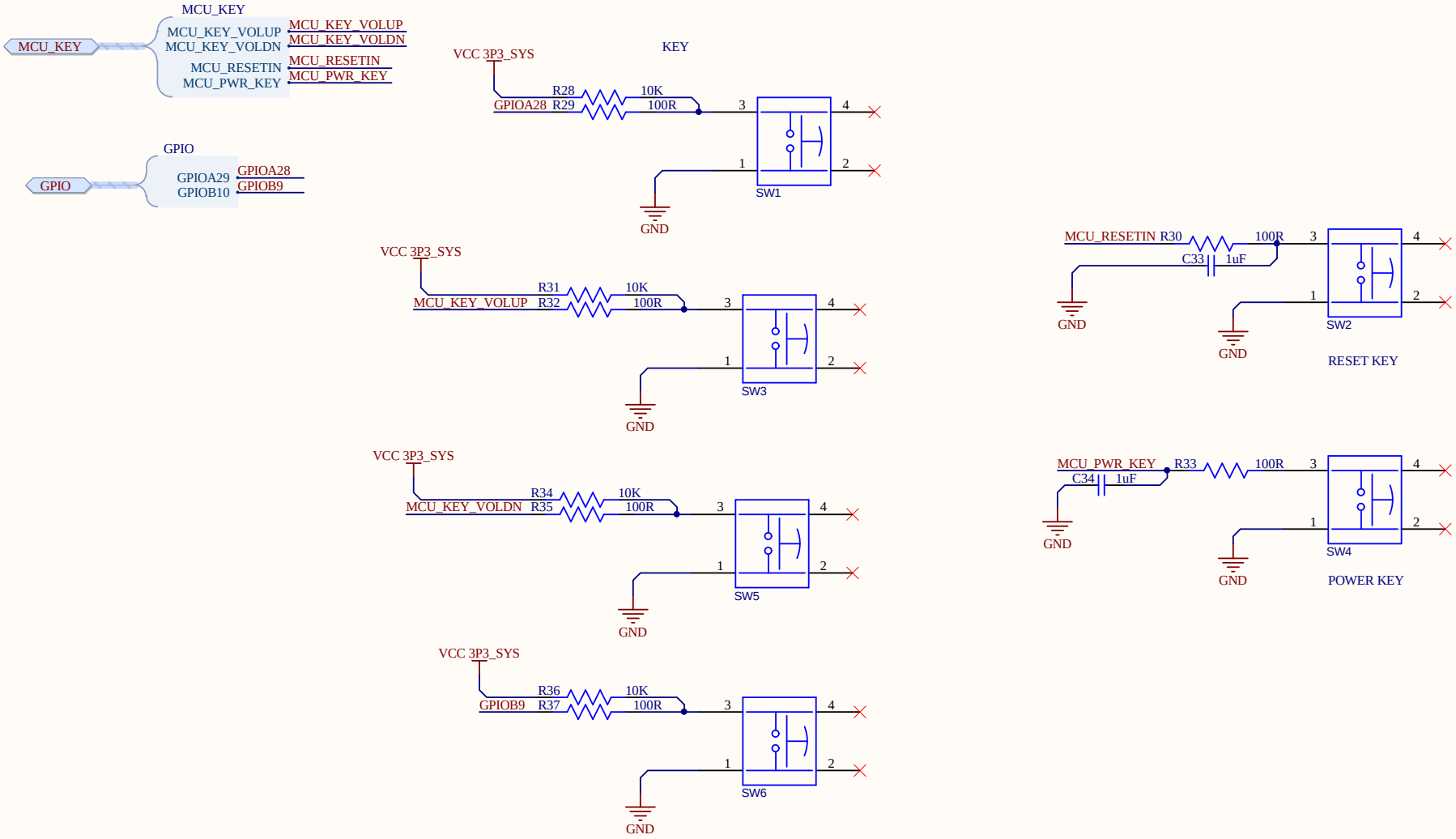

VBAT_USB
J14
1 MH1
HUB_DM4 2
HUB_DP4 3
4 MH2
87583-2010BLF
GND


SENSOR_I2C
SENSOR_I2C
MCU_KEY
MCU_KEY
GPIO
GPIO


U_Sensor_Section
Sensor_Section.SchDoc
VCC 3P3_SYS
VCC 3P3_SYS
SENSOR_I2C
SENSOR_I2C
GND
GND
GND

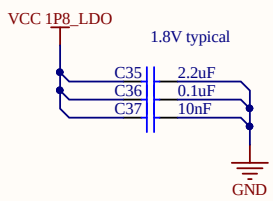

U_Key_Buttons
Key_Buttons.SchDoc
VCC 3P3_SYS
VCC 3P3_SYS
MCU_KEY
GPIO
MCU_KEY
GPIO
GND
GND

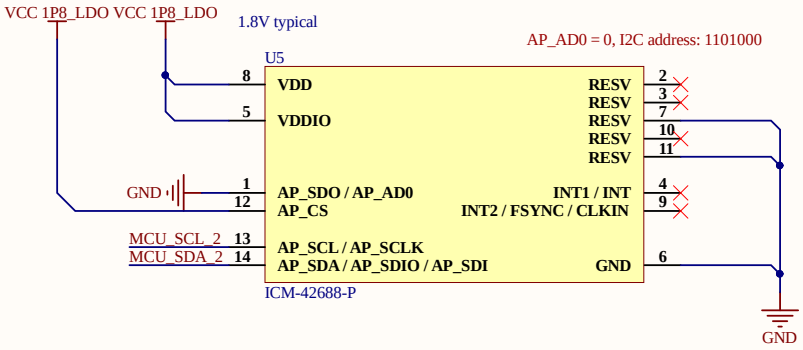

VCC 3P3_SYS
VCC 3P3_SYS
GND
GND

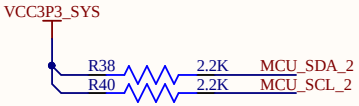

MCU_KEY
MCU_KEY
MCU_KEY_VOLUP
MCU_KEY_VOLDN
MCU_RESETIN
MCU_PWR_KEY
MCU_KEY_VOLUP
MCU_KEY_VOLDN
MCU_RESETIN
MCU_PWR_KEY
GPIO
GPIO
GPIOA29
GPIOB10
GPIOA28
GPIOB9
KEY
VCC 3P3_SYS
R28
10K
GPIOA28
R29
100R
SW1
GND
VCC 3P3_SYS
R31
10K
MCU_KEY_VOLUP
R32
100R
SW3
GND
VCC 3P3_SYS
R34
10K
MCU_KEY_VOLDN
R35
100R
SW5
GND
VCC 3P3_SYS
R36
10K
GPIOB9
R37
100R
SW6
GND
MCU_RESETIN
R30
100R
C33
1uF
SW2
GND
GND
RESET KEY
MCU_PWR_KEY
R33
100R
C34
1uF
SW4
GND
GND
POWER KEY
VCC 3P3_SYS
VCC 3P3_SYS
GND
GND

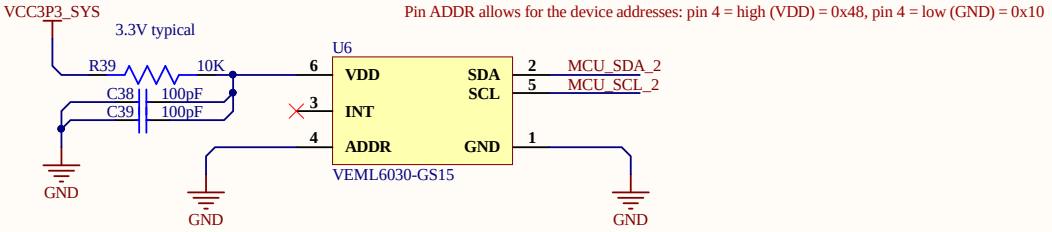
KEY BUTTONS

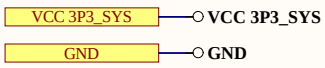



# SENSORS - GYRO, ACCEL & LIGHT